Research Methods

Collect & Analyze Finance Data from LSEG with Python (2025/2026)

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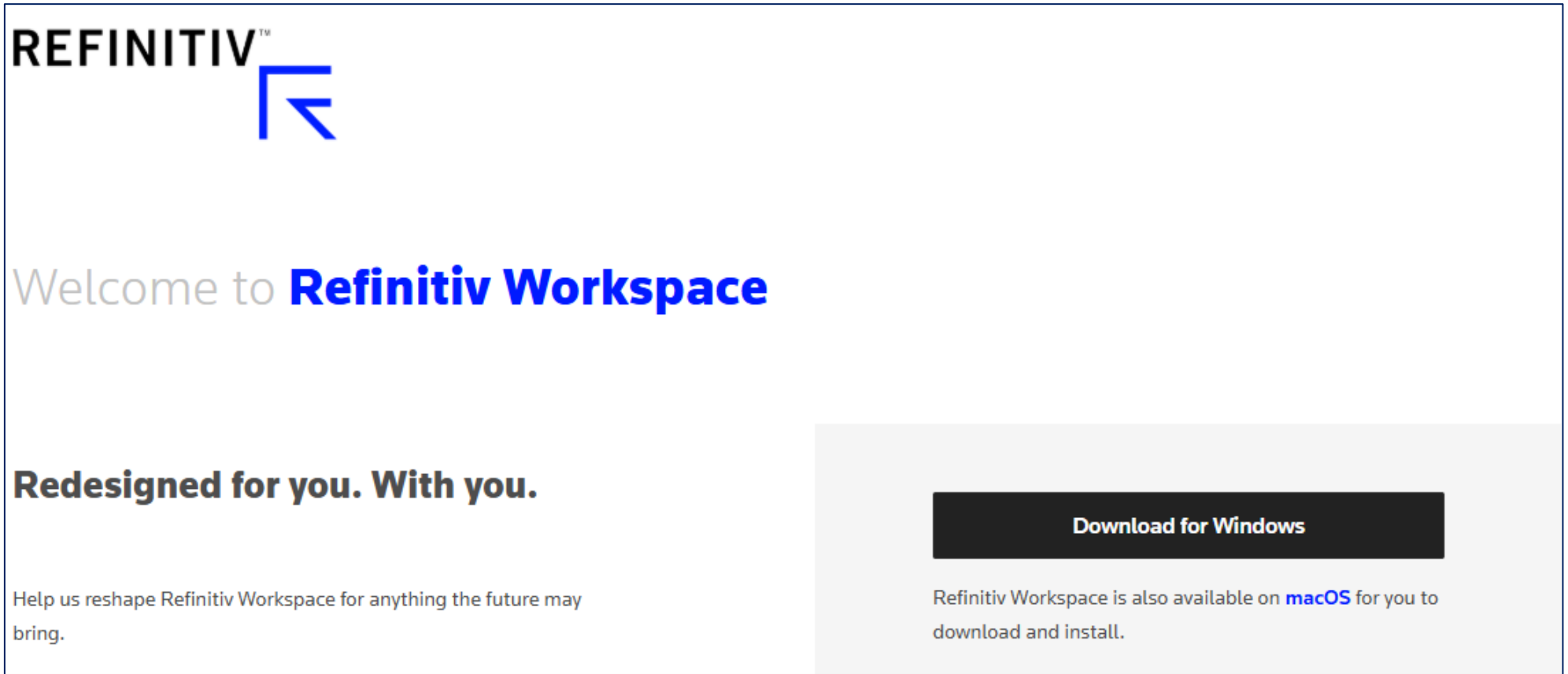
Summary

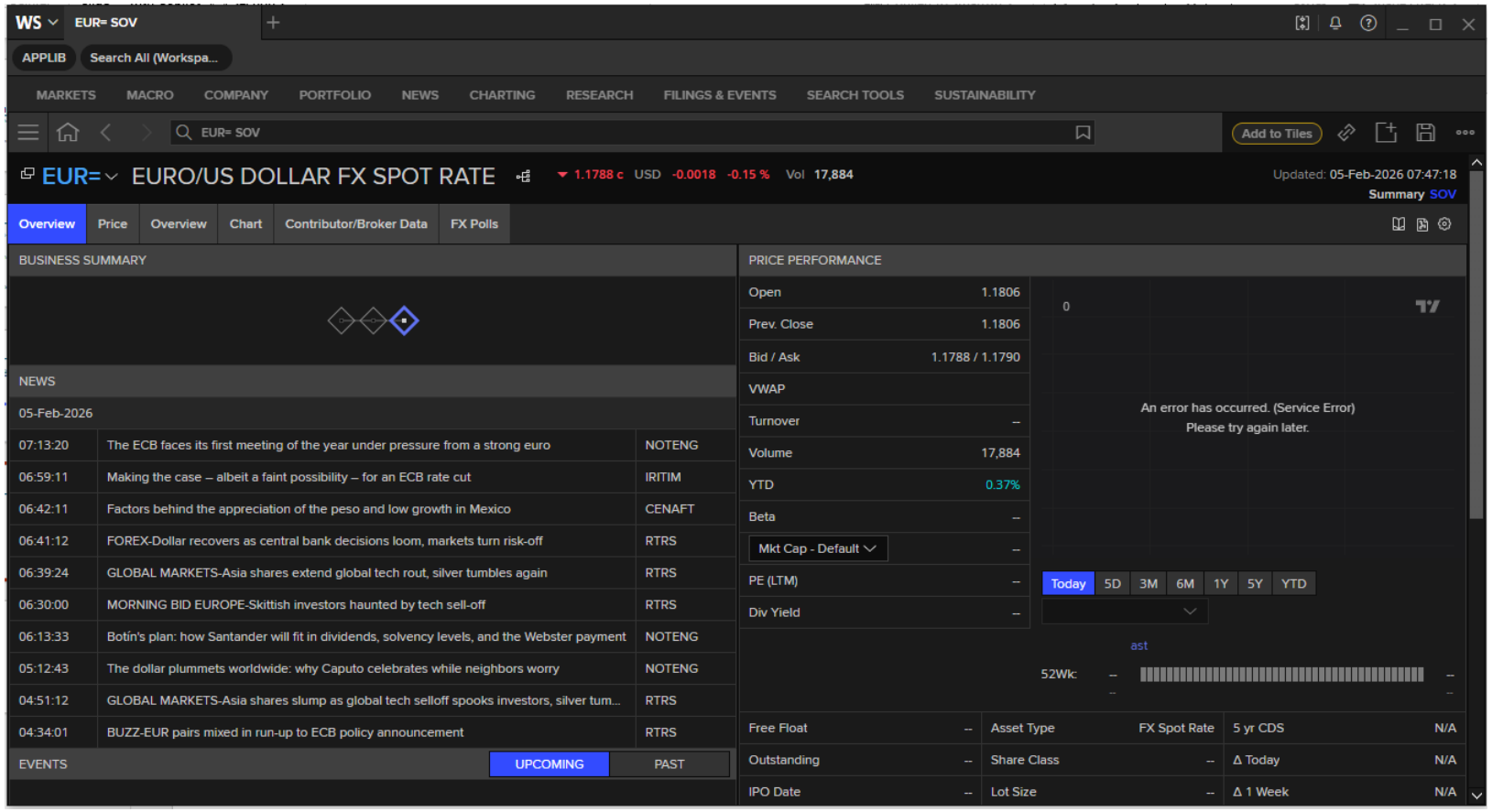
1. Refinitiv/LSEG
2. Collecting data with Python
3. Graph & charts with Python
4. Descriptive statistics with Python
5. T-test with Python
6. Regression analysis with Python
7. Event study with Python
8. Portfolio analysis with Python
9. Cursor

1. Refinitiv/LSEG

Install Refinitiv **Workplace**:
https://cdn.refinitiv.com/Apps/ProductDownloadPage/1.1.13/



1. Refinitiv/LSEG



Knowledge Hub sent you your login and password, shared among several students

Cannot have more than one student using a shared account simultaneously

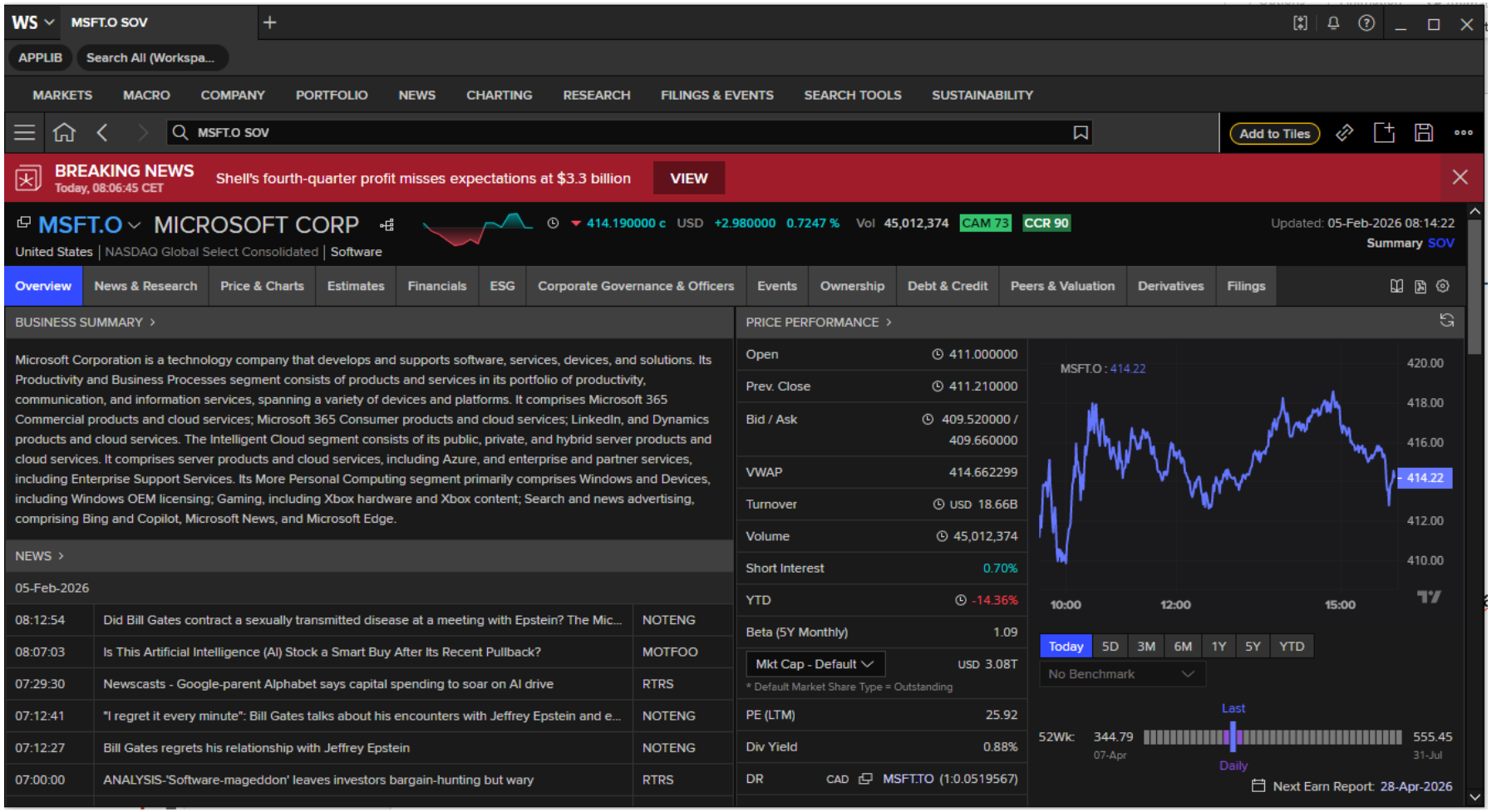
Hence, **collect and save the data** for later running the analyses

1. Refinitiv/LSEG

Available data:

- Accounting (book value, sales, capex, R&D, earnings)
- Financials (bond, stock, option)
- Forecasts (EPS, BPS, price)
- Ownership
- ESG (data points, scores, controversies)
- Filings (annual reports, earning calls, integrated reports)
- Fund information
- Country information
- News flow
- Prices (commodities, FOREX)
- M&A, Deals of PE

1. Refinitiv/LSEG



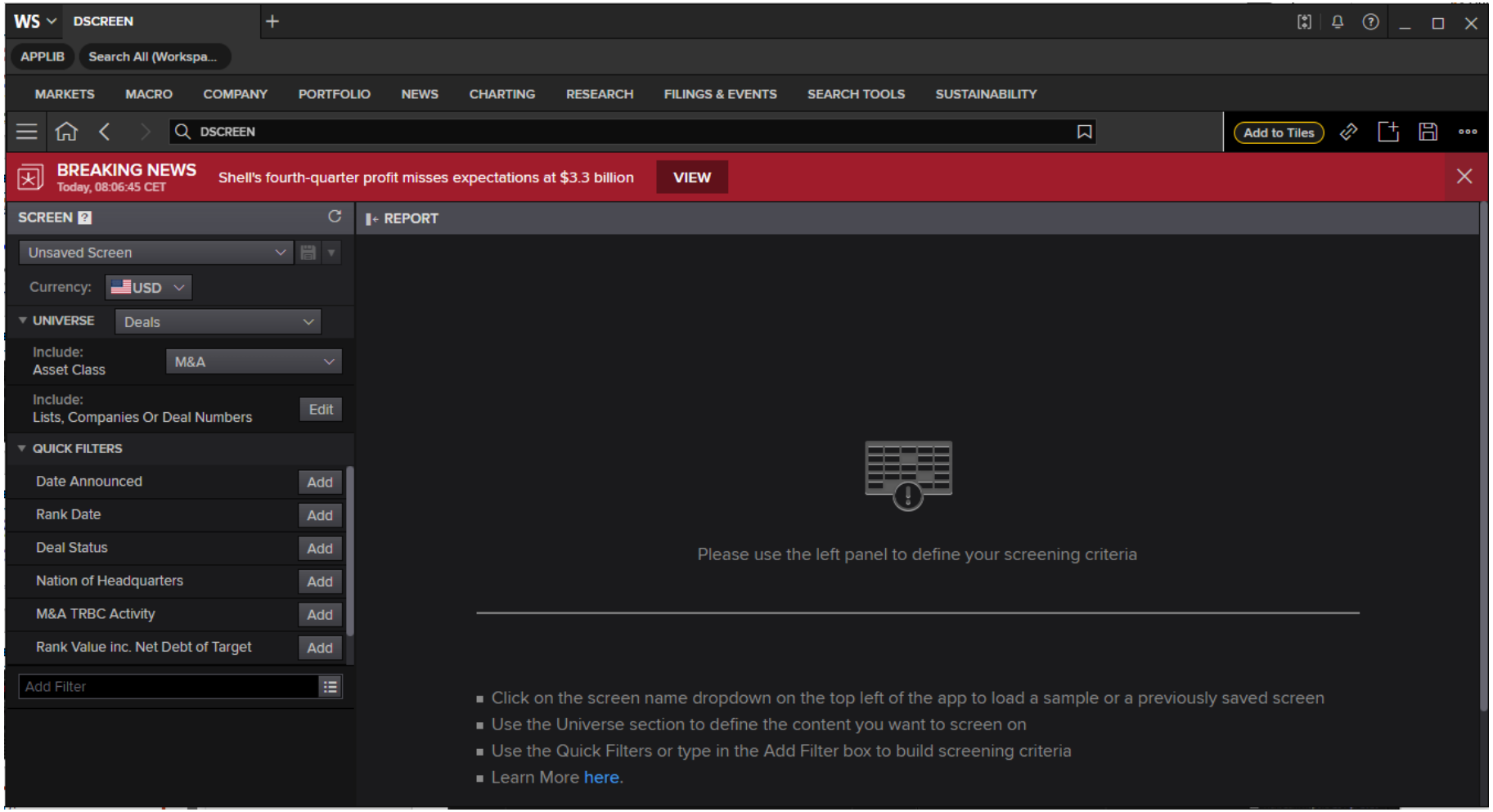
We search for **Microsoft**

We type Microsoft in the search bar or use its **RIC** (secutity identifier)

Also works with *ISIN* or *CUSIP*, other security identifiers

Let us explore the content together

1. Refinitiv/LSEG



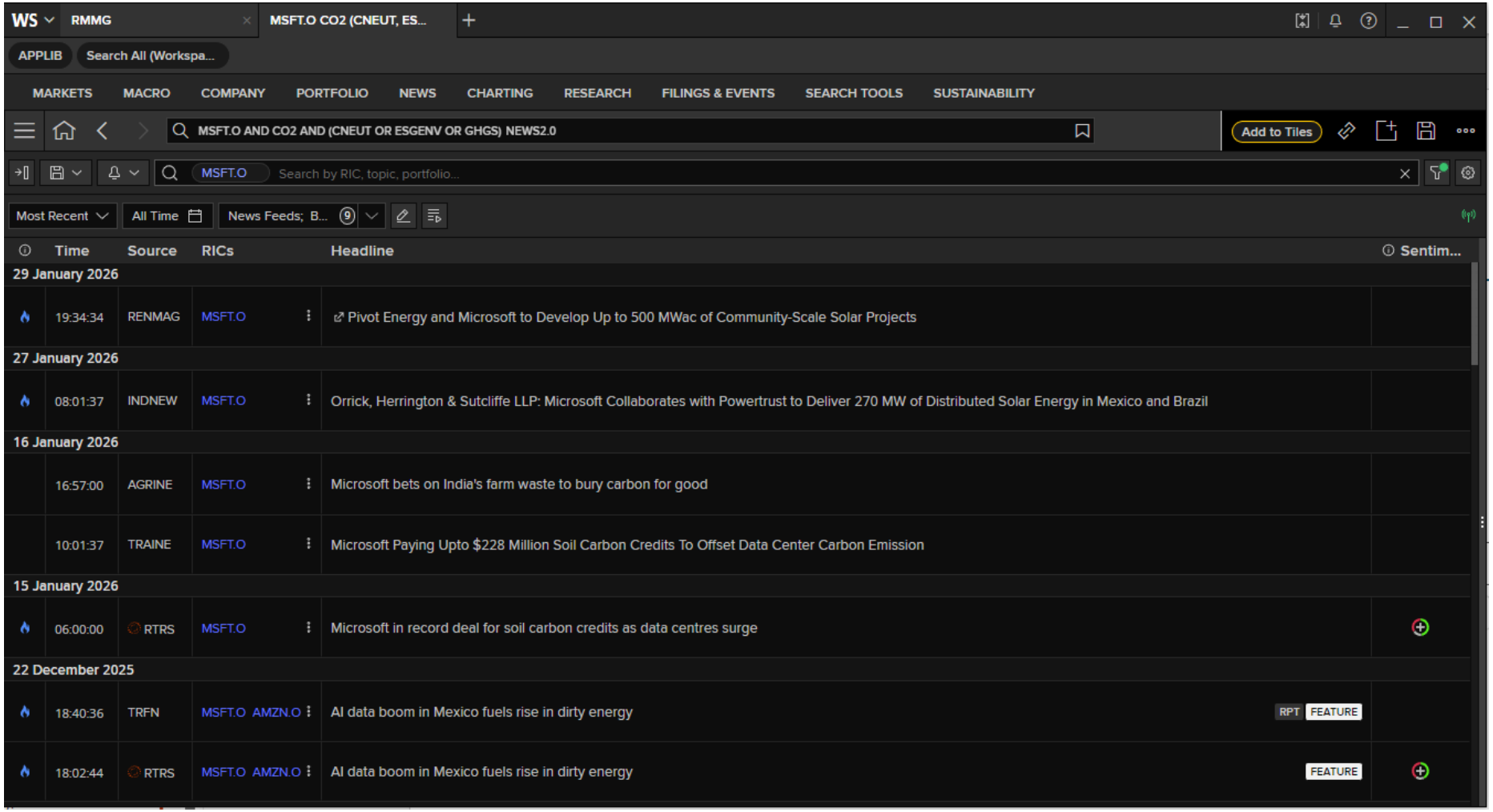
Screeners

Companies

Deals screener (bond, stock, M&A, PE)

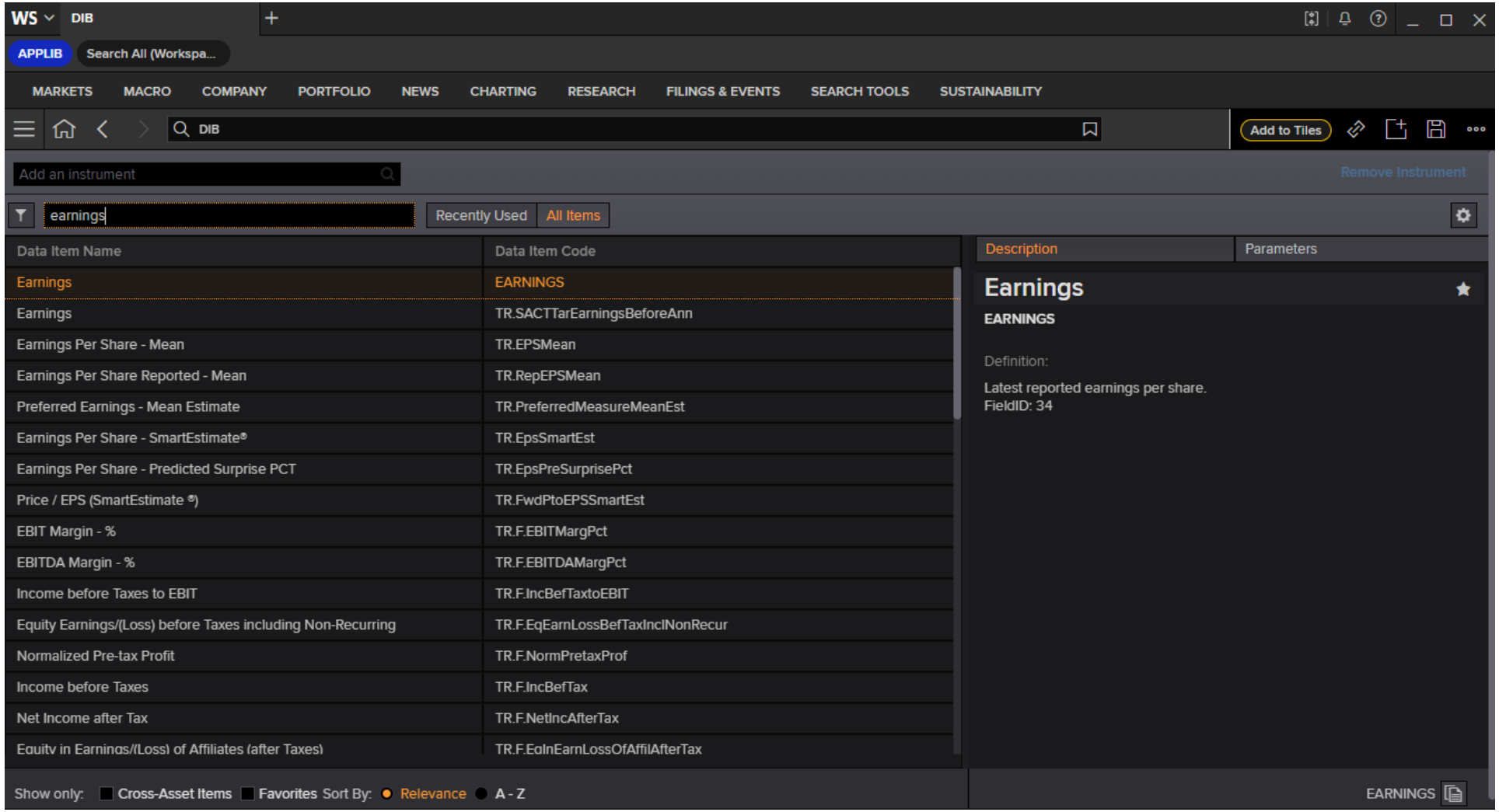
Fund screener

1. Refinitiv/LSEG



News monitor

1. Refinitiv/LSEG



Data Item Browser

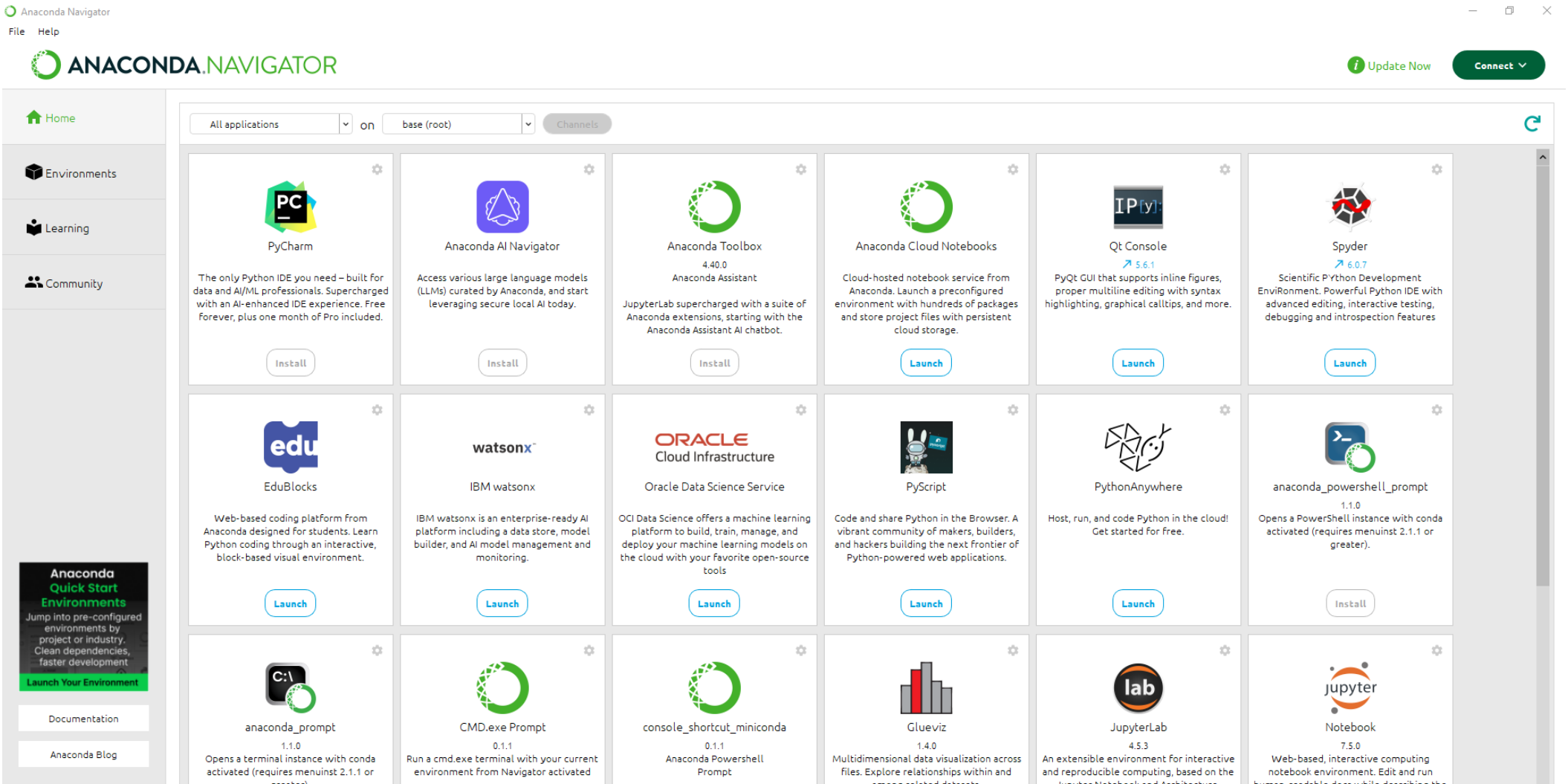
2. Collecting data with Python

- Why **Python**?

- **Anaconda** bundles Python, Jupyter notebooks, and common data packages in one installer.
  - Installs Python automatically
  - Works on both Mac and Windows

- We download and install it : http://anaconda.com/download

2. Collecting data with Python

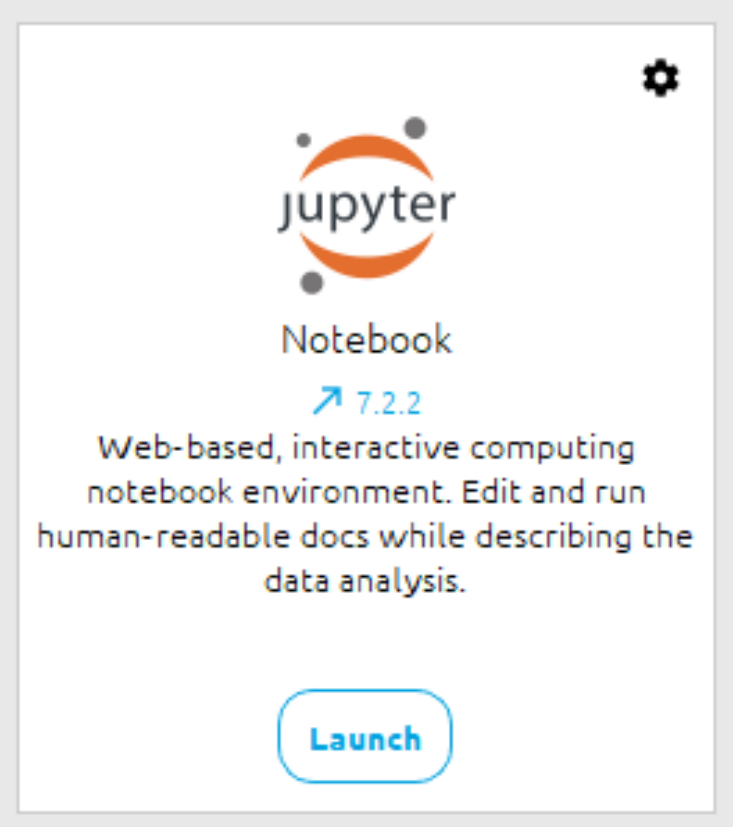


2. Collecting data with Python

- Install/launch **Jupyter Notebook**

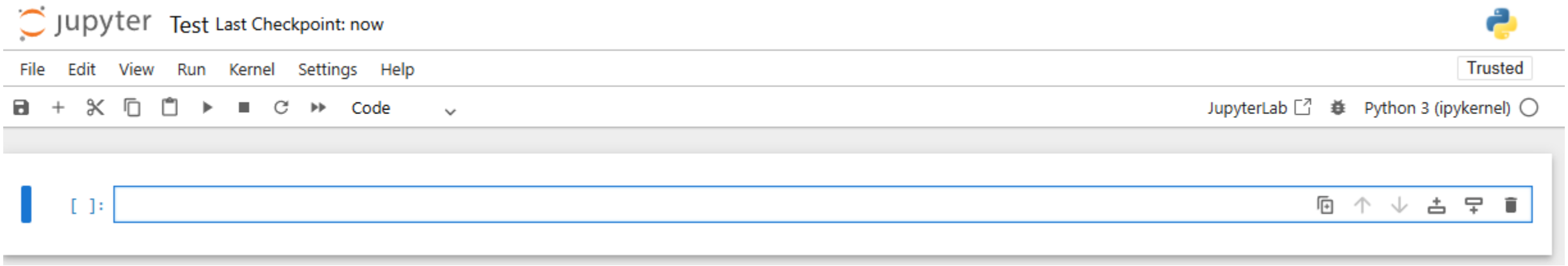
- Jupyter Notebook is a tool that lets you write code, run it, and see the results all in one place—inside your web browser.

- Allows to run code step by step/ see output immediately (numbers, tables, plots) / add text explanations next to the code.



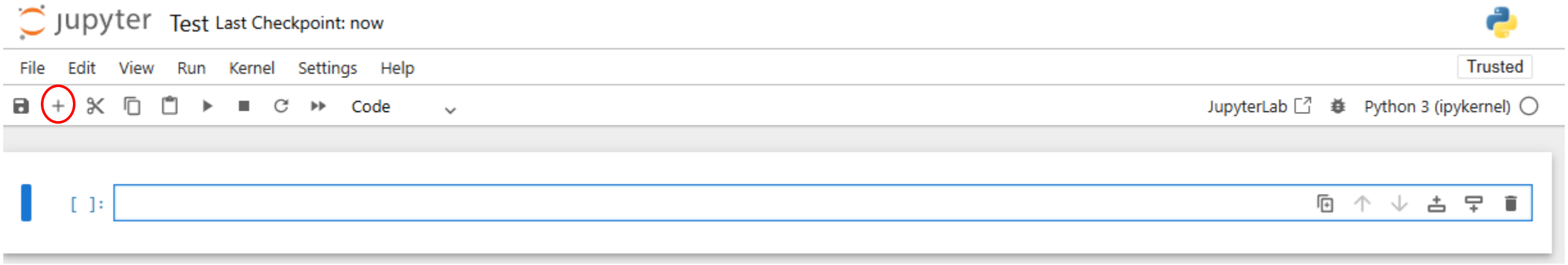
2. Collecting data with Python

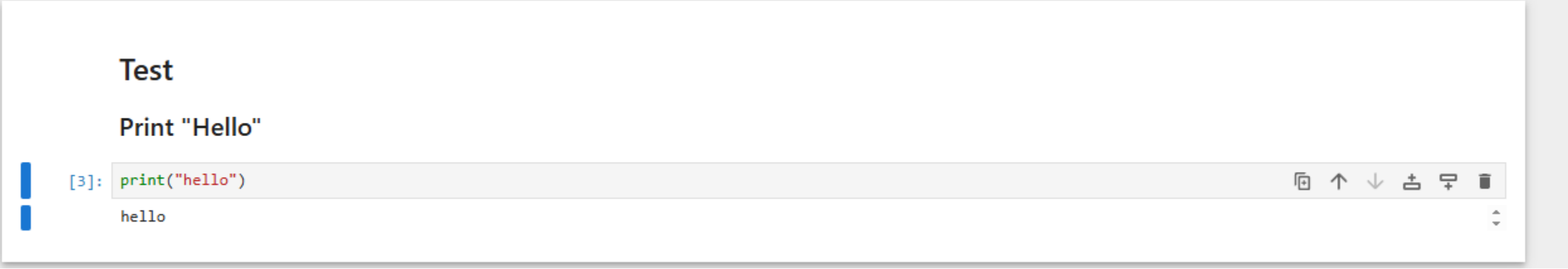
- Launch Jupyter Notebook
- Choose folder
- File > New > Notebook (Python 3)
- Save it as *Test.ipynb*



2. Collecting data with Python

➢ With the + button, we can add text (markdown) or code section (python), and them execute





2. Collecting data with Python

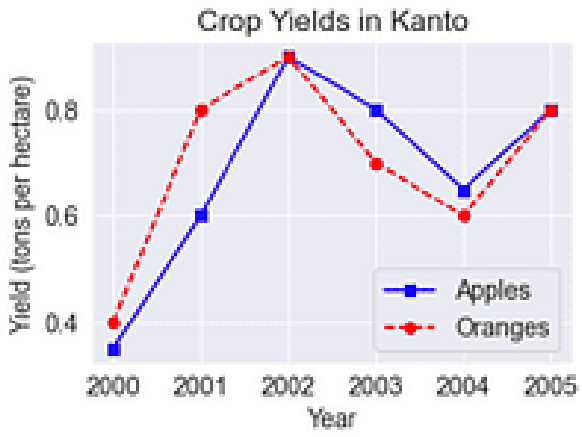
We switch to the **research_method_notebook**

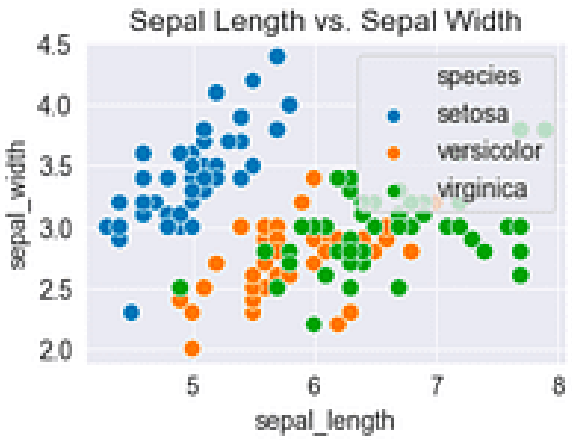
All the material, including this presentation is available on **Blackboard**.

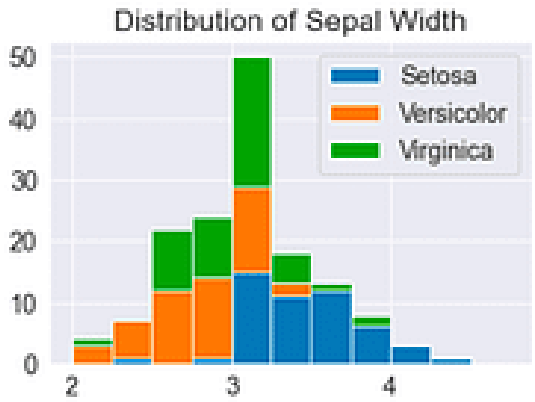
2. Collecting data with Python

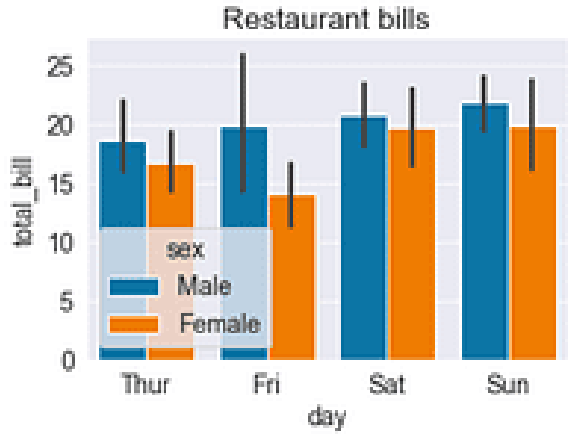
Now that we have reviewed the part of the **research_method_notebook** dedicated to data collection, try to collect datasets possibly relevant for your own master thesis...

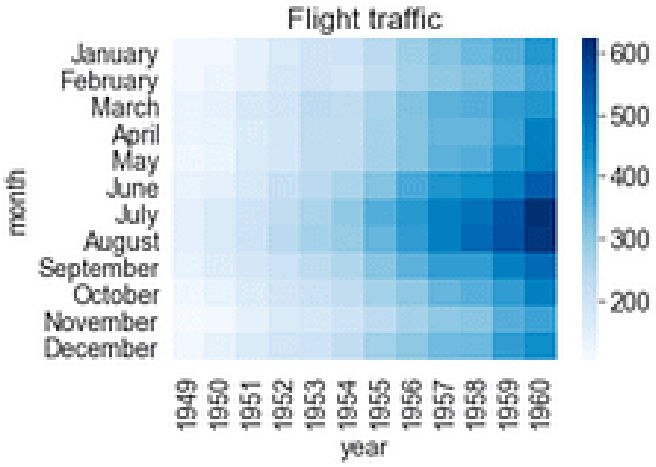
3. Data visualization in Python

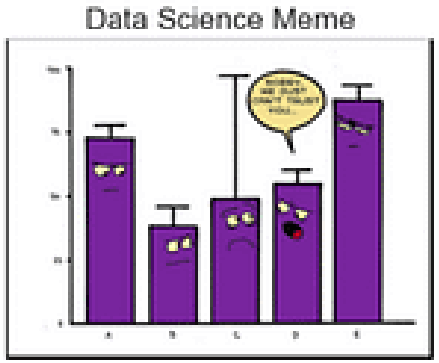




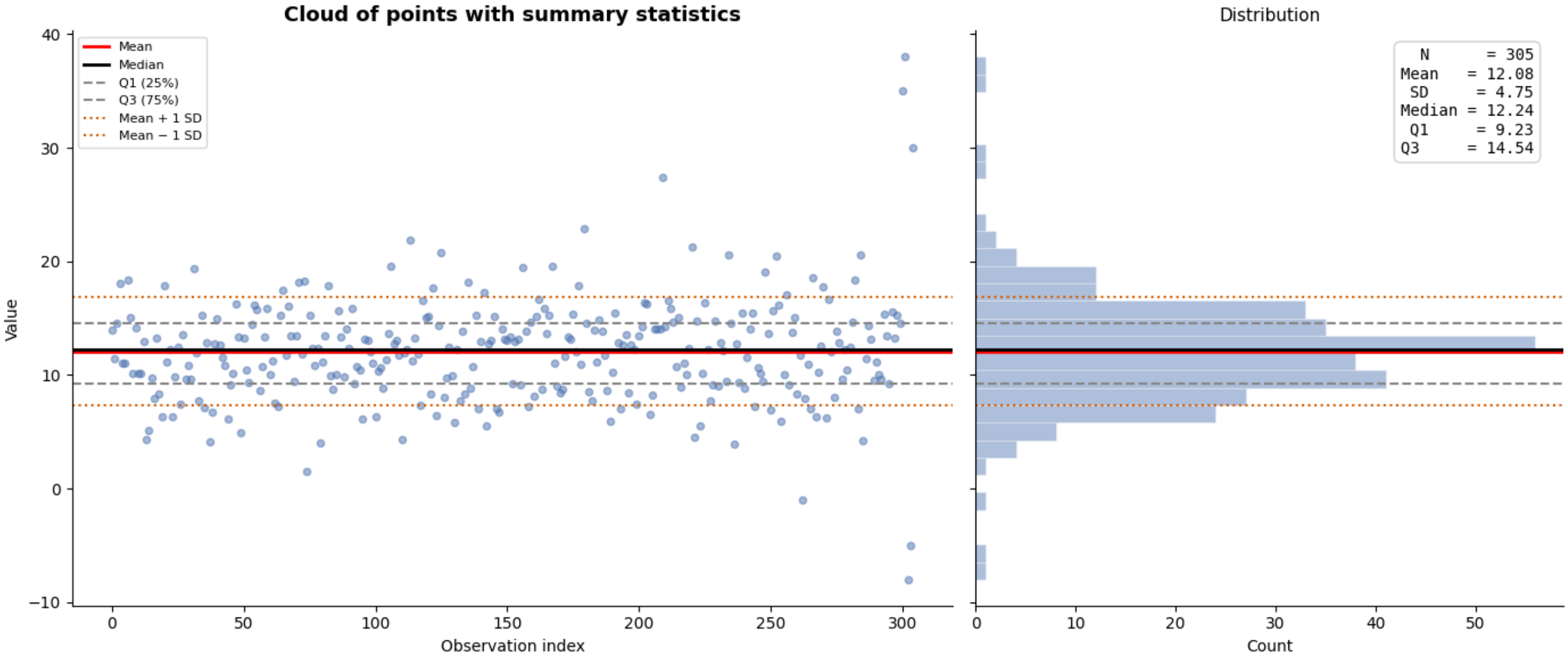




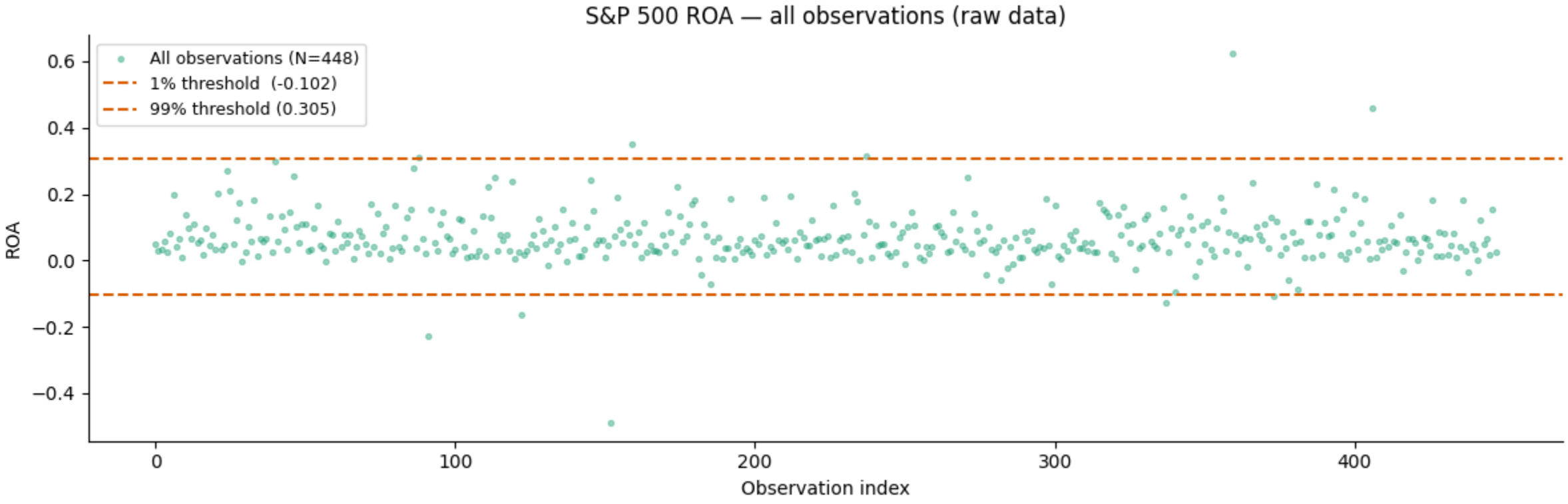




4. Descriptive statistics with Python



4. Winsorizing/Trimming



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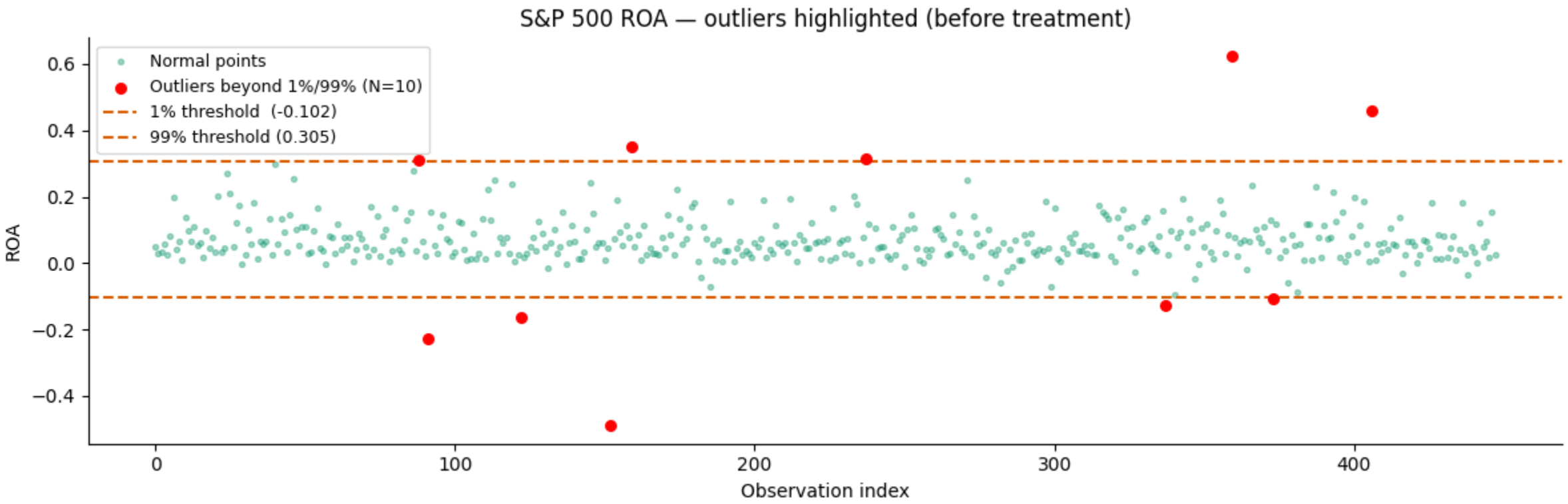

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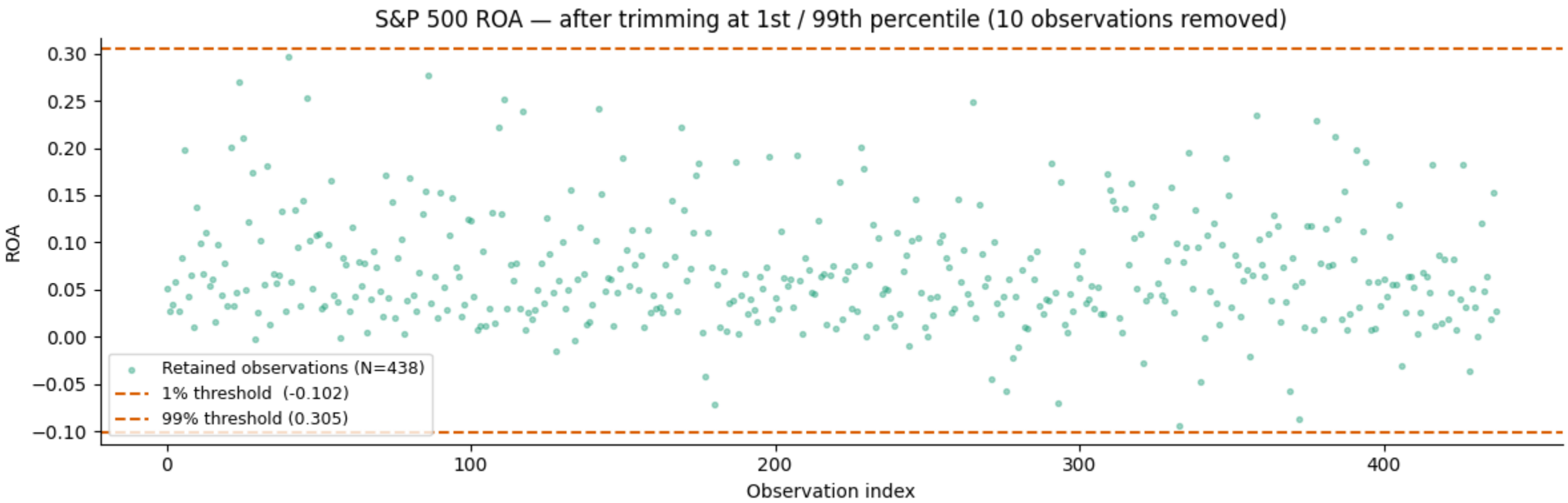

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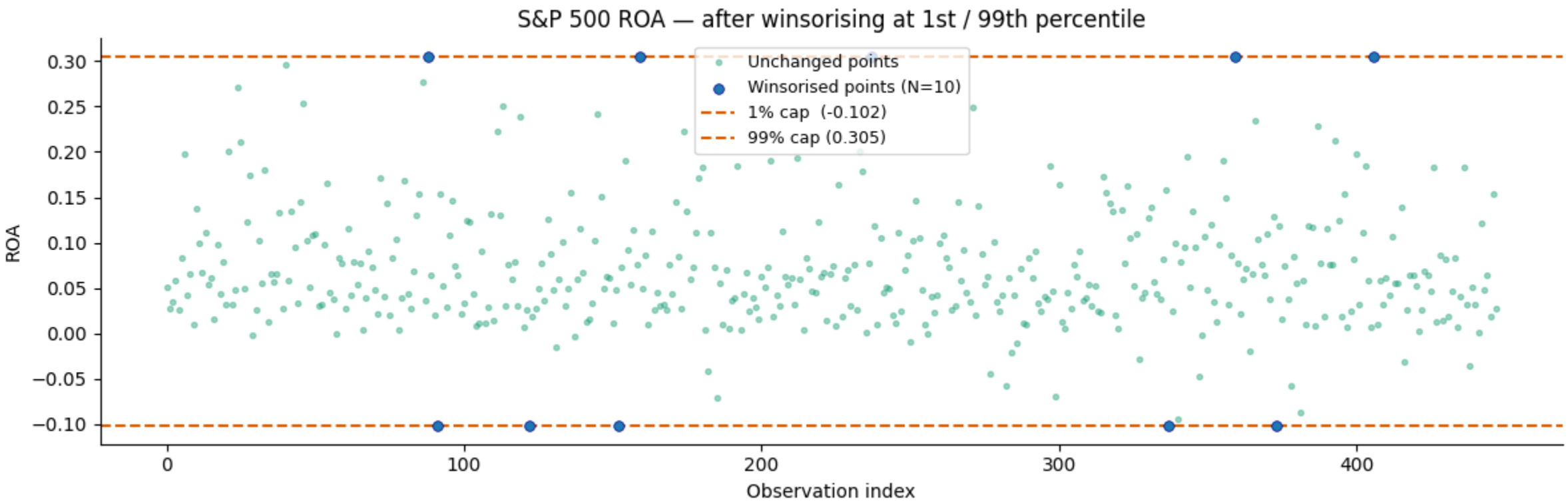

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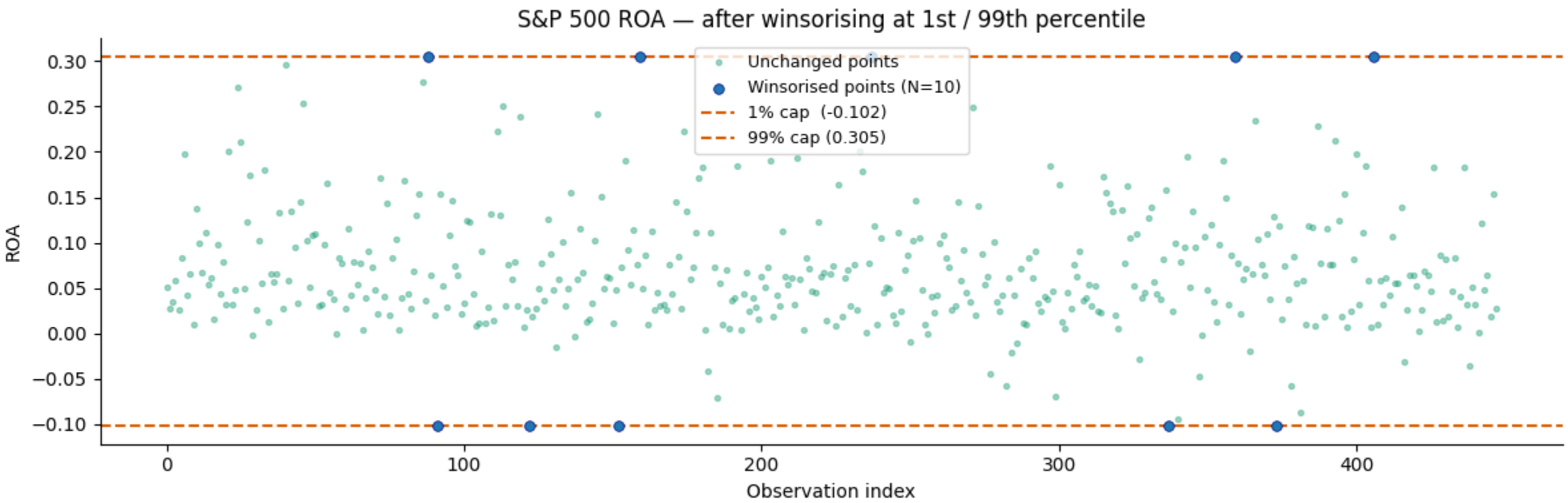

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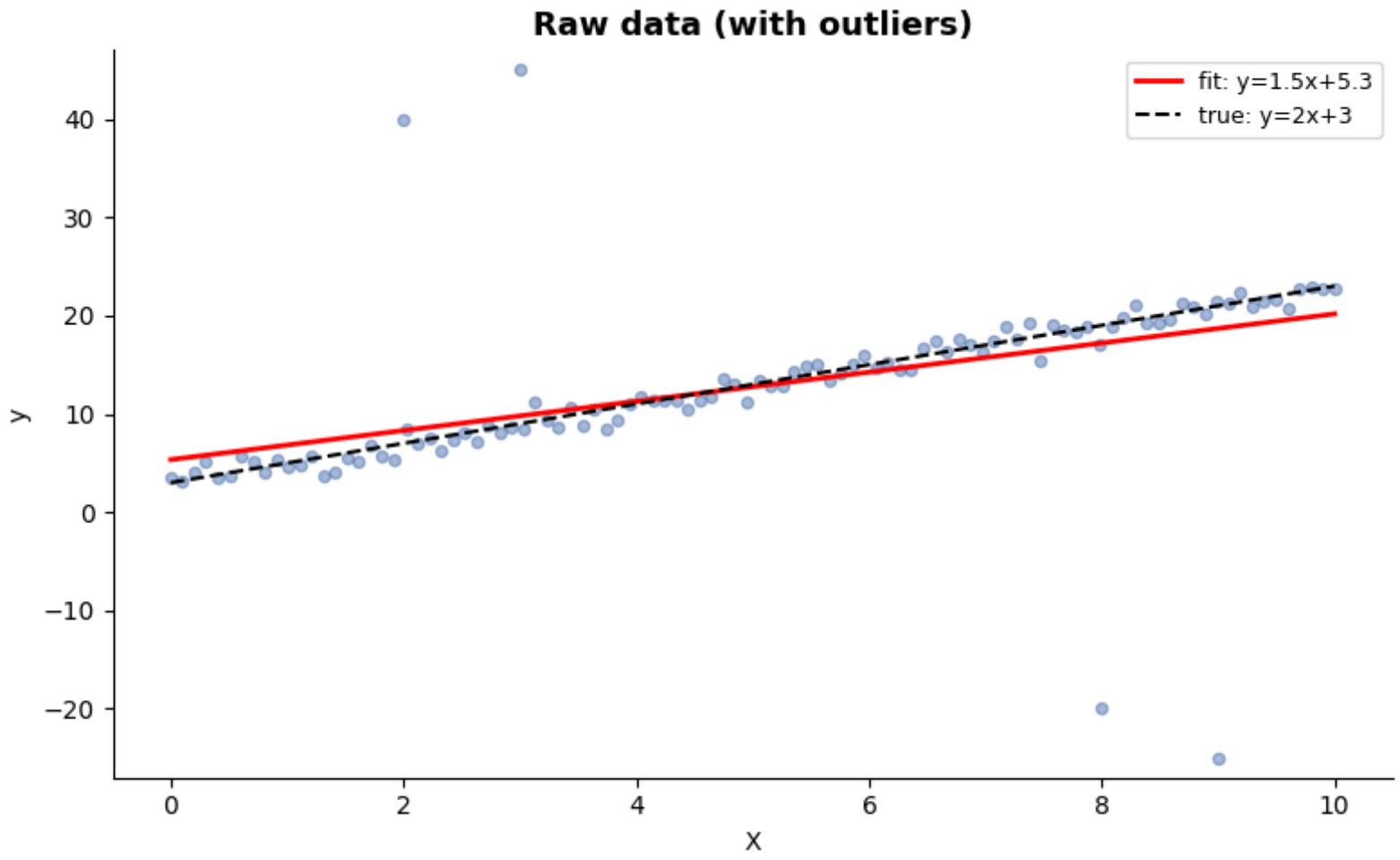

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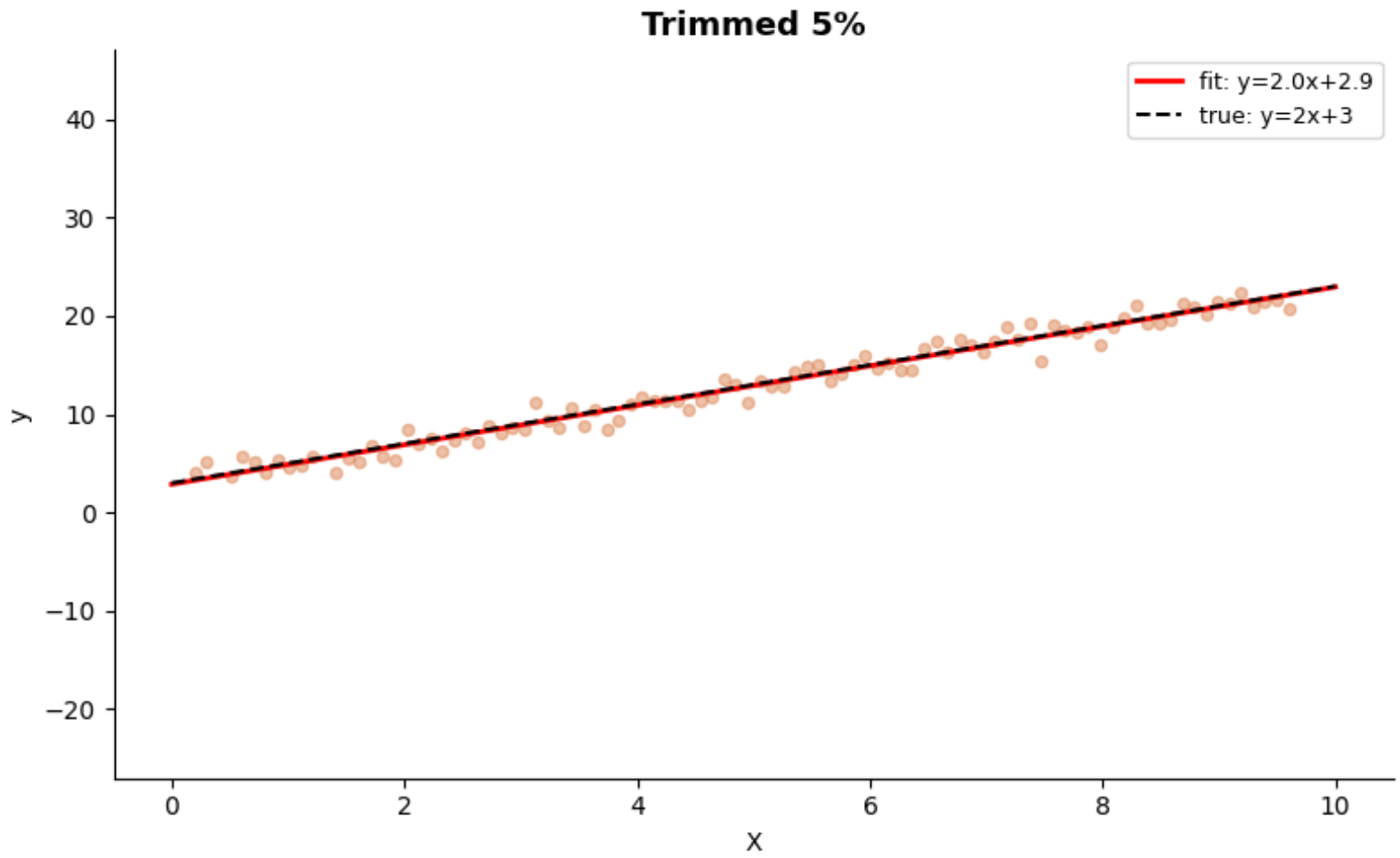


5. T-test

A **population** is the entire set of individuals, firms, or observations you want to draw conclusions about (e.g. all listed firms in a given year). It's the "truth" — but in practice it's usually too large or impossible to observe in full.

A **sample** is the subset you actually collect and analyse. Because it's only a piece of the population,

5. T-test

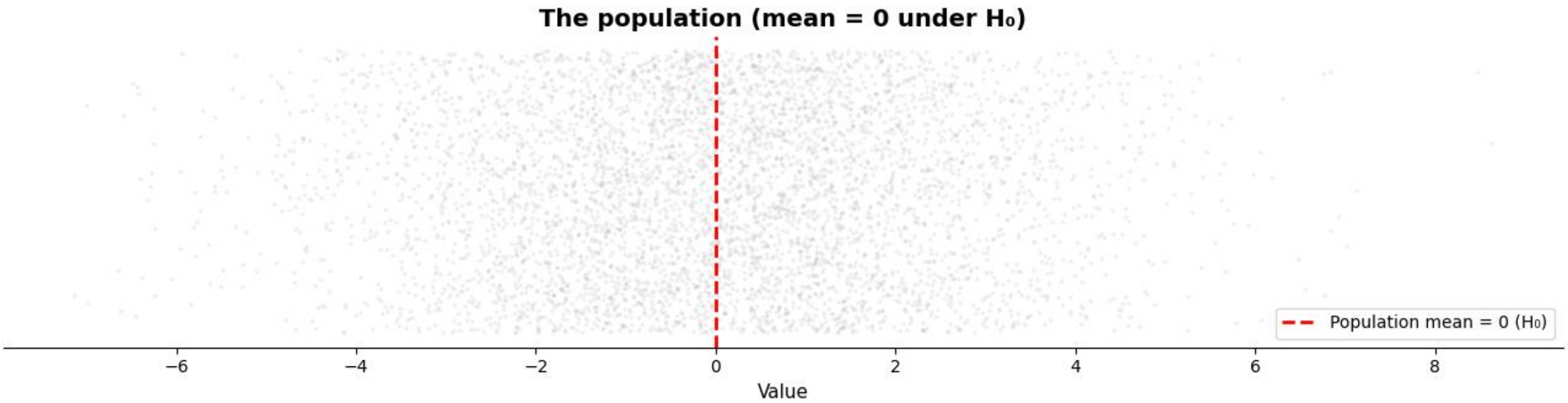
A **population** is the entire set of individuals, firms, or observations you want to draw conclusions about (e.g. all listed firms in a given year). It's the "truth" — but in practice it's usually too large or impossible to observe in full.

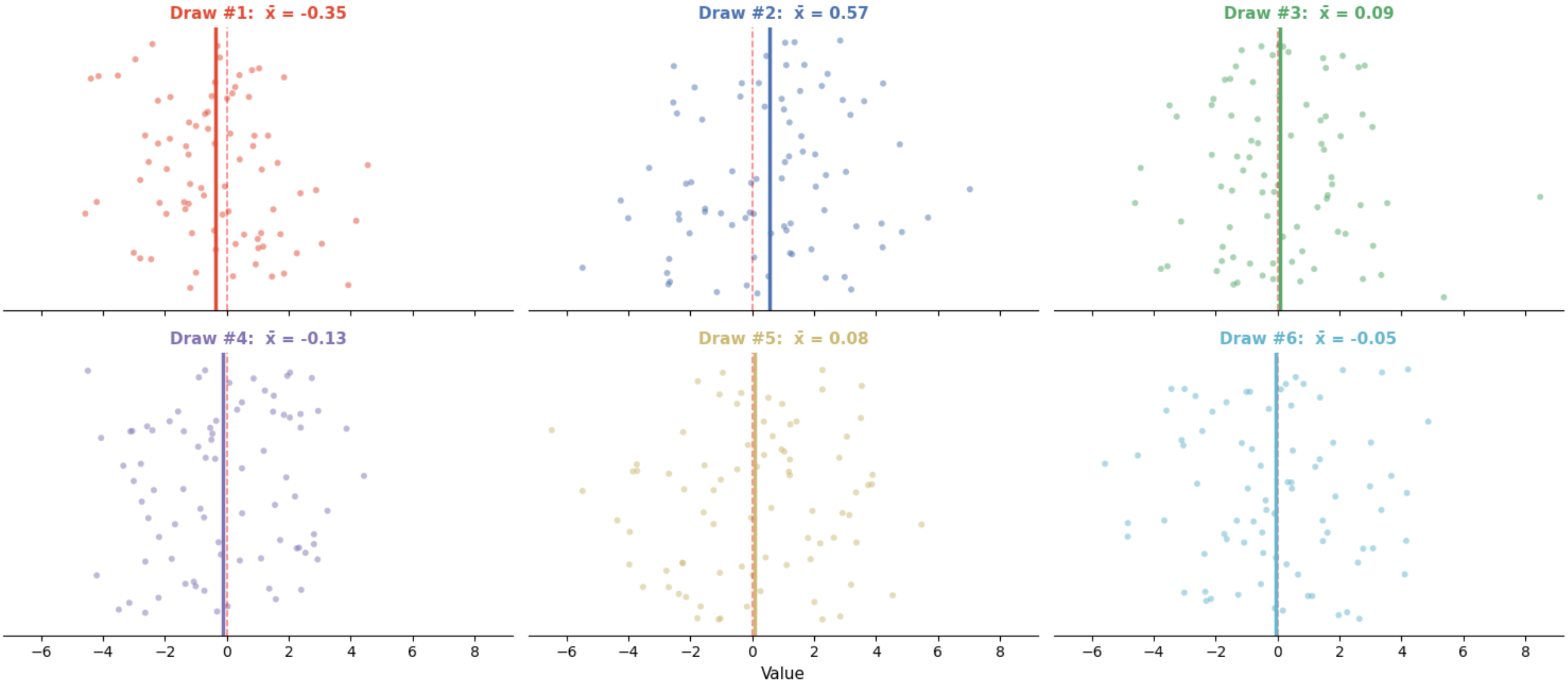
A **sample** is the subset you actually collect and analyse. Because it's only a piece of the population,

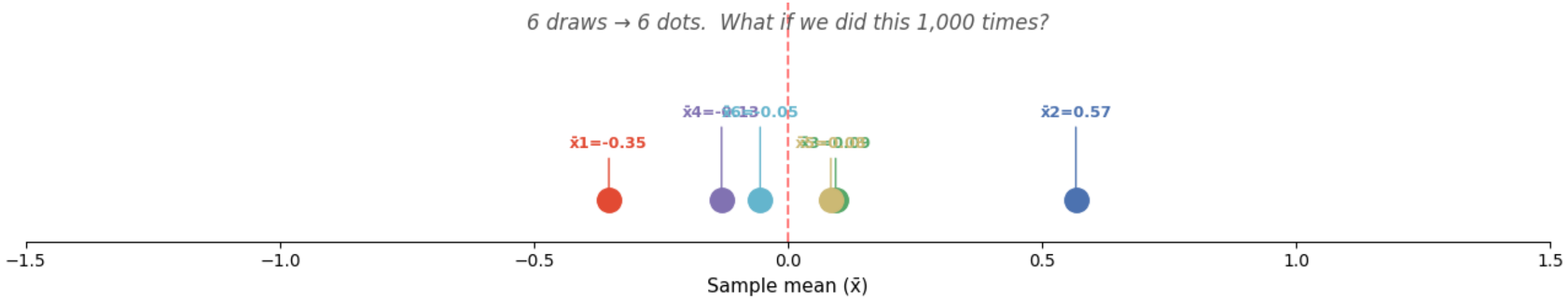
5. T-test

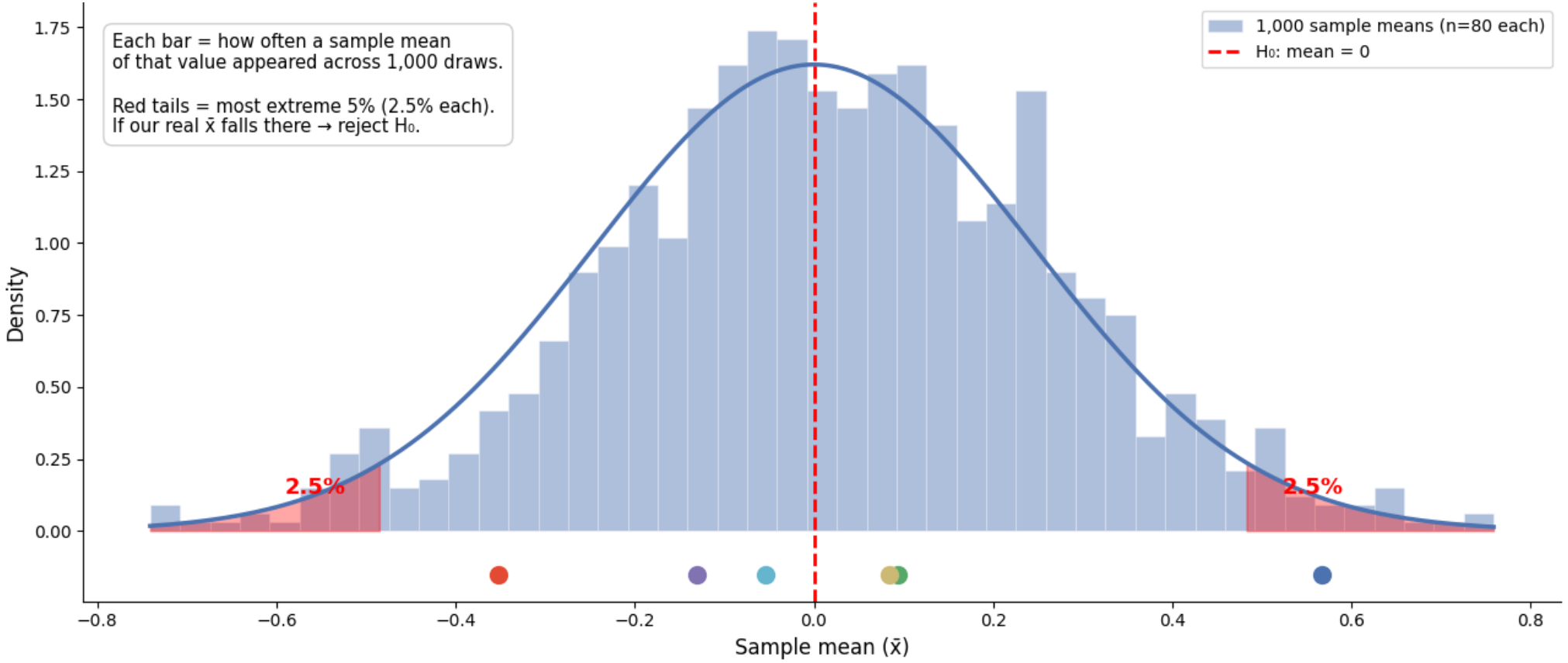
Null hypothesis that the mean is zero in the population (H0). We generate a **p-value** to reject/not reject the hypothesis.

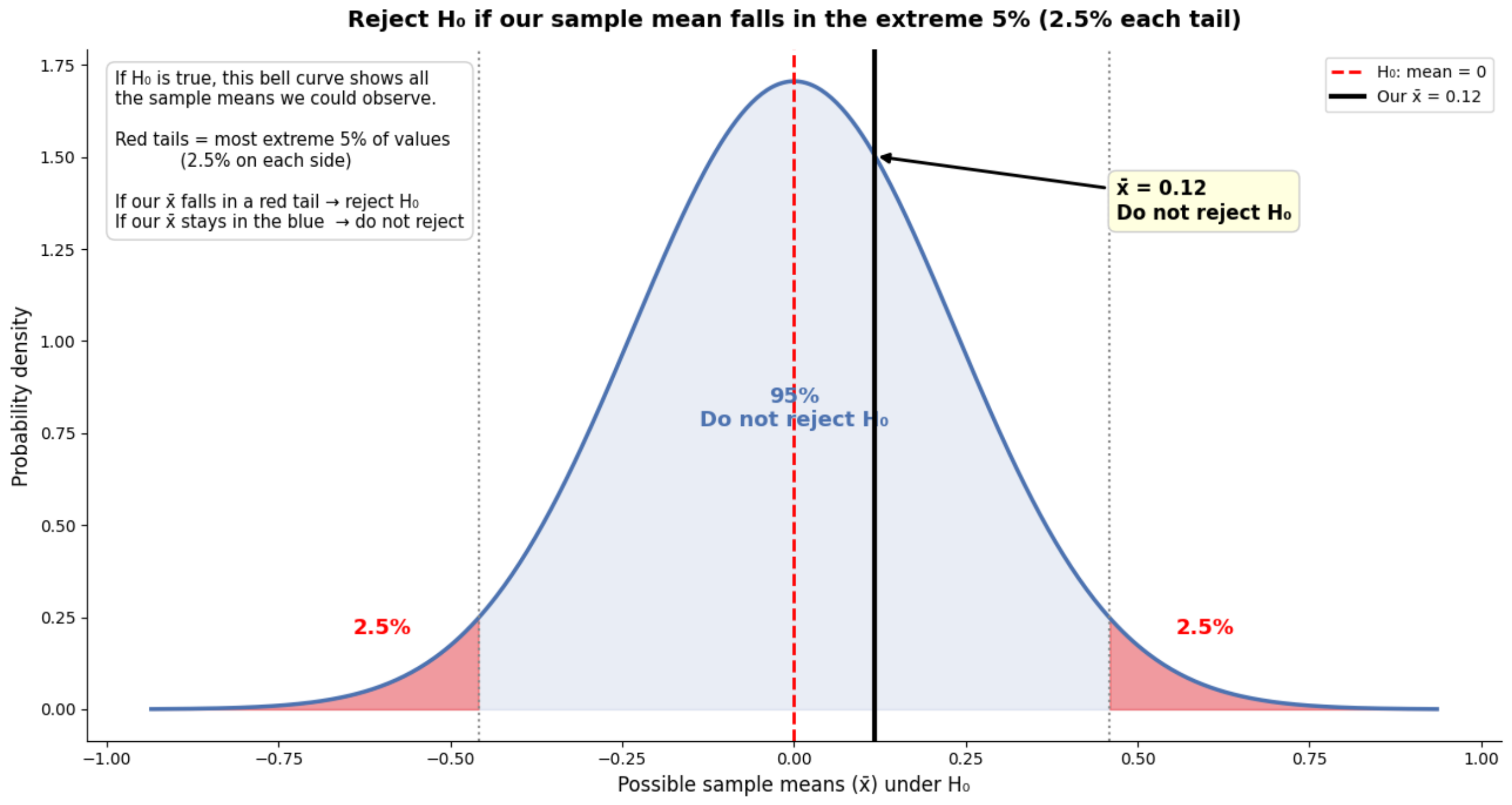
The **p-value** is the probability of observing a sample mean as extreme as ours (or more extreme) if H_0 were actually true.

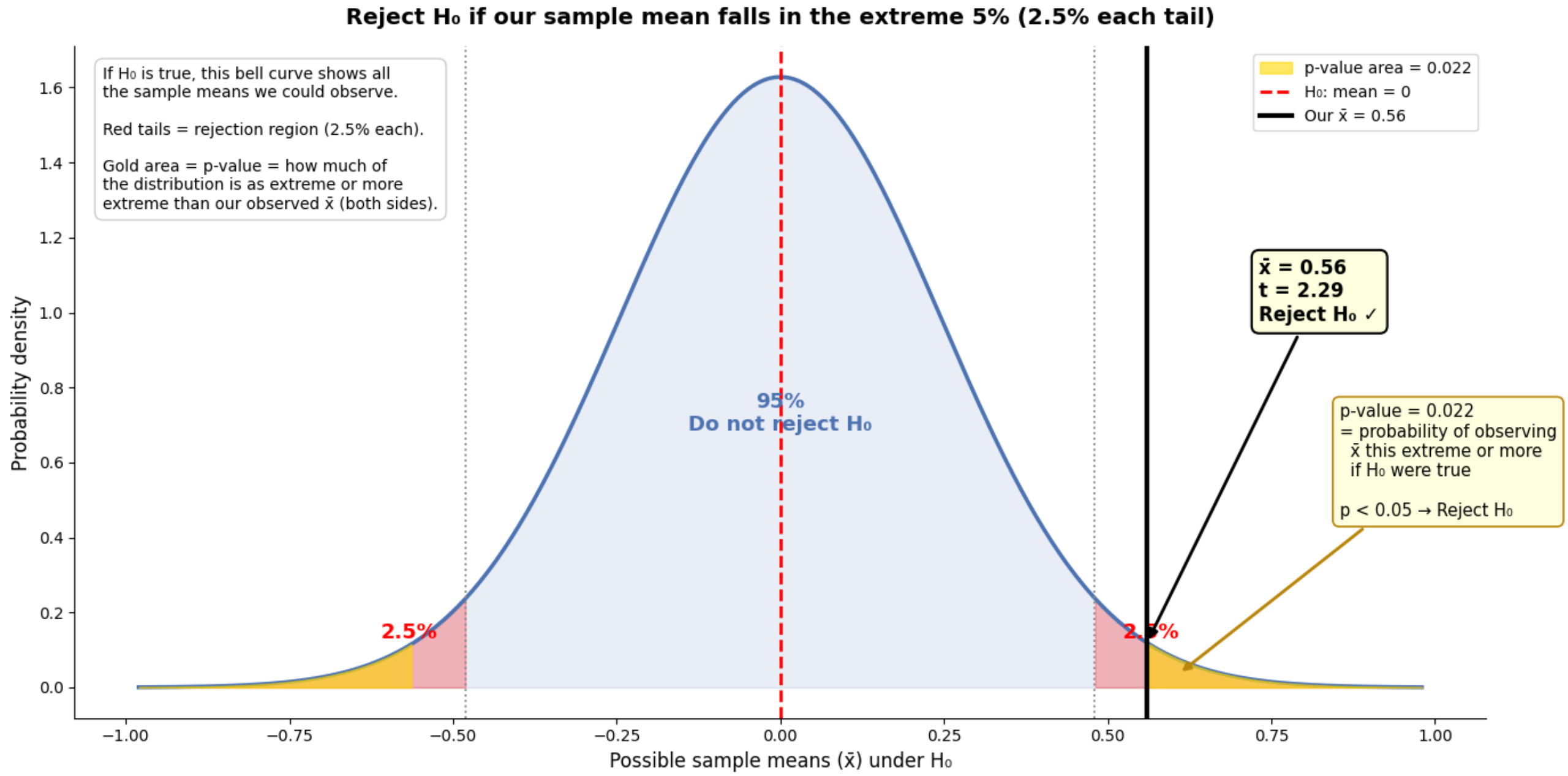


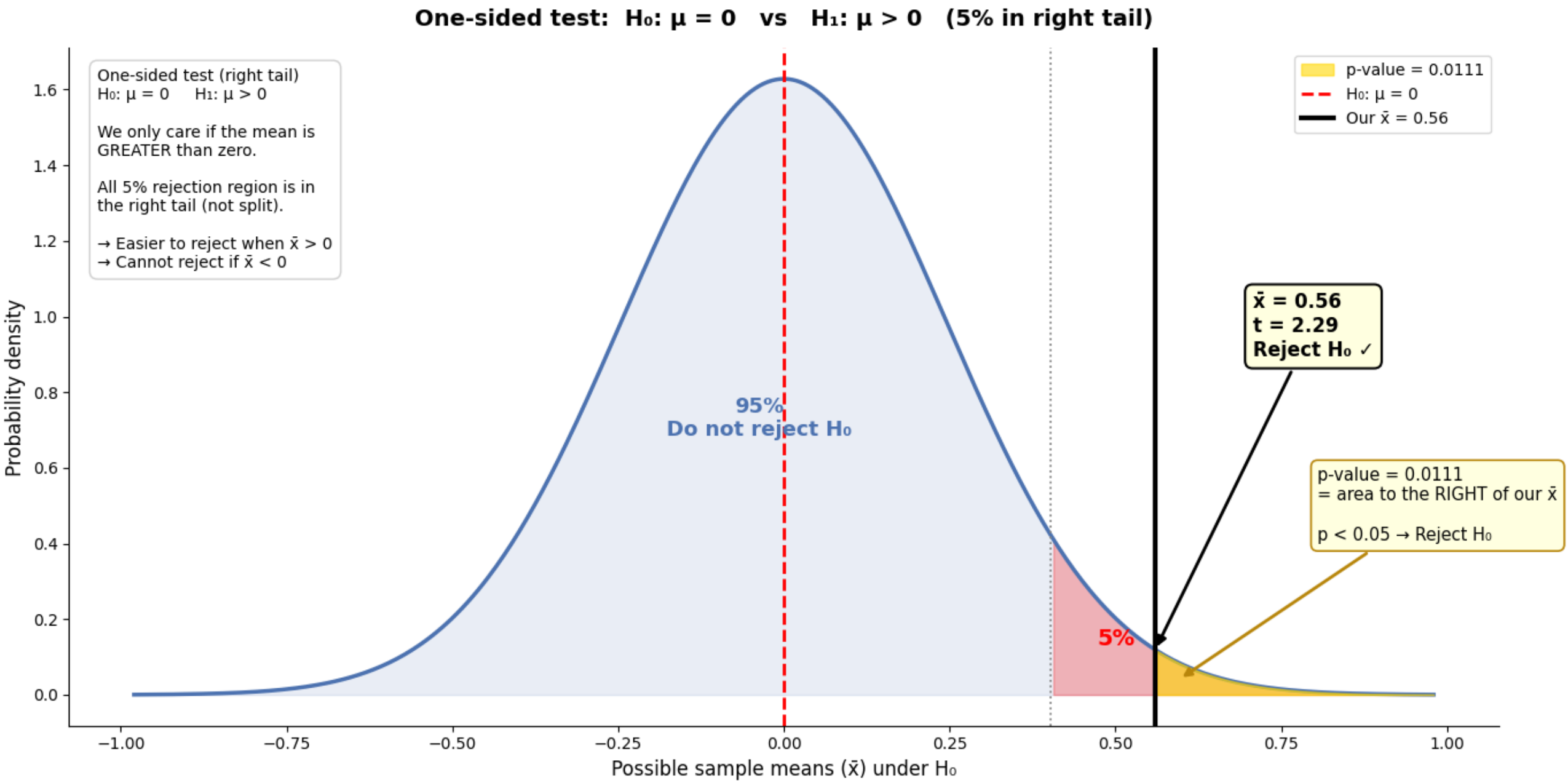

Draw #1: x̄ = -0.35
Draw #2: x̄ = 0.57
Draw #3: x̄ = 0.09
Draw #4: x̄ = -0.13
Draw #5: x̄ = 0.08
Draw #6: x̄ = -0.05
−6 −4 −2 0 2 4 6 8
Value

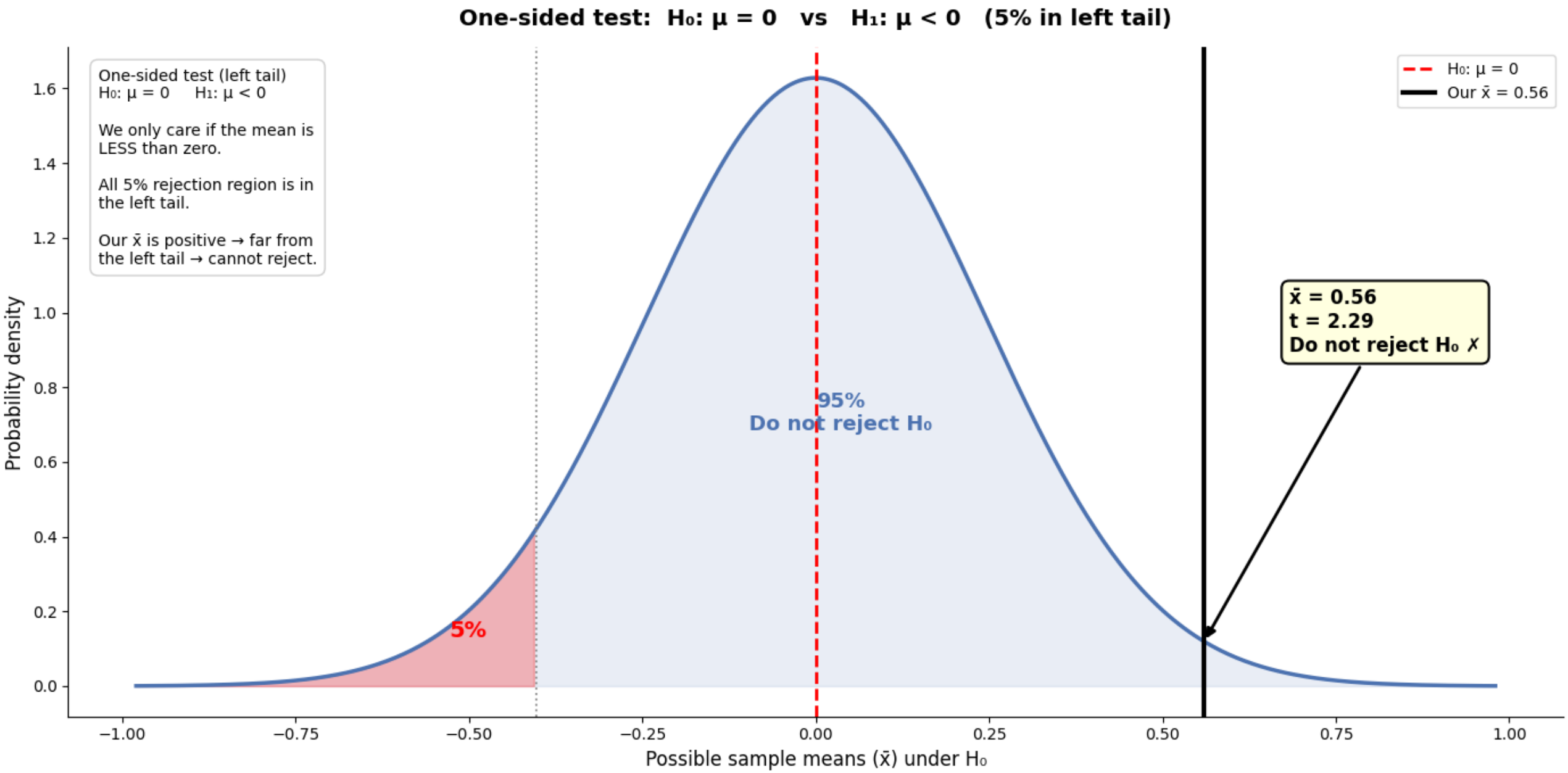

6 draws → 6 dots. What if we did this 1,000 times?
x̄1=-0.35
x̄2=0.57
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Sample mean (x̄)


Each bar = how often a sample mean
of that value appeared across 1,000 draws.
Red tails = most extreme 5% (2.5% each).
If our real x̄ falls there → reject H₀.
1,000 sample means (n=80 each)
H₀: mean = 0
2.5%
2.5%
Density
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
Sample mean (x̄)


Reject H₀ if our sample mean falls in the extreme 5% (2.5% each tail)
If H₀ is true, this bell curve shows all the sample means we could observe.
Red tails = most extreme 5% of values (2.5% on each side)
If our x̄ falls in a red tail → reject H₀
If our x̄ stays in the blue → do not reject
H₀: mean = 0
Our x̄ = 0.12
x̄ = 0.12
Do not reject H₀
95%
Do not reject H₀
2.5%
2.5%
Probability density
Possible sample means (x̄) under H₀
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00


Reject H₀ if our sample mean falls in the extreme 5% (2.5% each tail)
If H₀ is true, this bell curve shows all the sample means we could observe.
Red tails = rejection region (2.5% each).
Gold area = p-value = how much of the distribution is as extreme or more extreme than our observed x̄ (both sides).
p-value area = 0.022
H₀: mean = 0
Our x̄ = 0.56
x̄ = 0.56
t = 2.29
Reject H₀ ✓
p-value = 0.022
= probability of observing x̄ this extreme or more if H₀ were true
p < 0.05 → Reject H₀
95%
Do not reject H₀
2.5%
2.5%
Probability density
Possible sample means (x̄) under H₀
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00


One-sided test: H₀: μ = 0 vs H₁: μ > 0 (5% in right tail)
One-sided test (right tail)
H₀: μ = 0 H₁: μ > 0
We only care if the mean is GREATER than zero.
All 5% rejection region is in the right tail (not split).
→ Easier to reject when x̄ > 0
→ Cannot reject if x̄ < 0
p-value = 0.0111
H₀: μ = 0
Our x̄ = 0.56
x̄ = 0.56
t = 2.29
Reject H₀ ✓
95%
Do not reject H₀
5%
p-value = 0.0111
= area to the RIGHT of our x̄
p < 0.05 → Reject H₀
Probability density
Possible sample means (x̄) under H₀
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00


One-sided test: H₀: μ = 0 vs H₁: μ < 0 (5% in left tail)
One-sided test (left tail)
H₀: μ = 0 H₁: μ < 0
We only care if the mean is LESS than zero.
All 5% rejection region is in the left tail.
Our x̄ is positive → far from the left tail → cannot reject.
H₀: μ = 0
Our x̄ = 0.56
x̄ = 0.56
t = 2.29
Do not reject H₀ ✗
95%
Do not reject H₀
5%
Probability density
Possible sample means (x̄) under H₀
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00

6. Portfolio analysis

What is a value-weighted portfolio?

Each firm's weight = its market cap ÷ total market cap of all firms

Total market cap = $1000B

Portfolio return $R_p = \sum w_i \times R_i$ = 50%×R_A + 30%×R_B + 15%×R_C + 5%×R_D

6. Portfolio analysis

Mean return, standard deviation & Sharpe ratio

Mean (μ)

Average return
over the period

$$\mu = \frac{1}{T}\sum_{t=1}^{T} R_t$$

"How much do we earn on average?"

Std Dev (σ)

Spread of returns
around the mean

$$\sigma = \sqrt{\frac{1}{T-1}\sum(R_t - \mu)^2}$$

"How bumpy is the ride?"

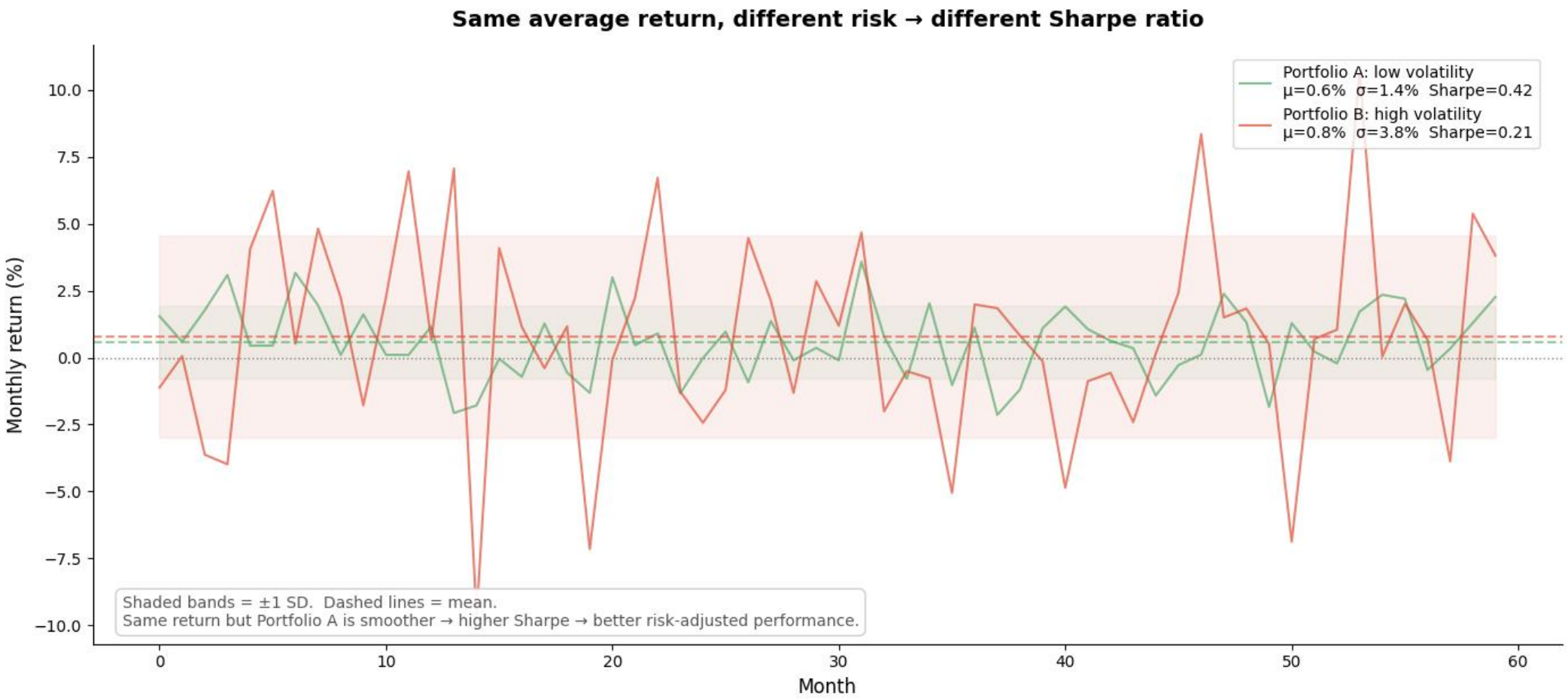
Sharpe Ratio

Return per unit
of risk

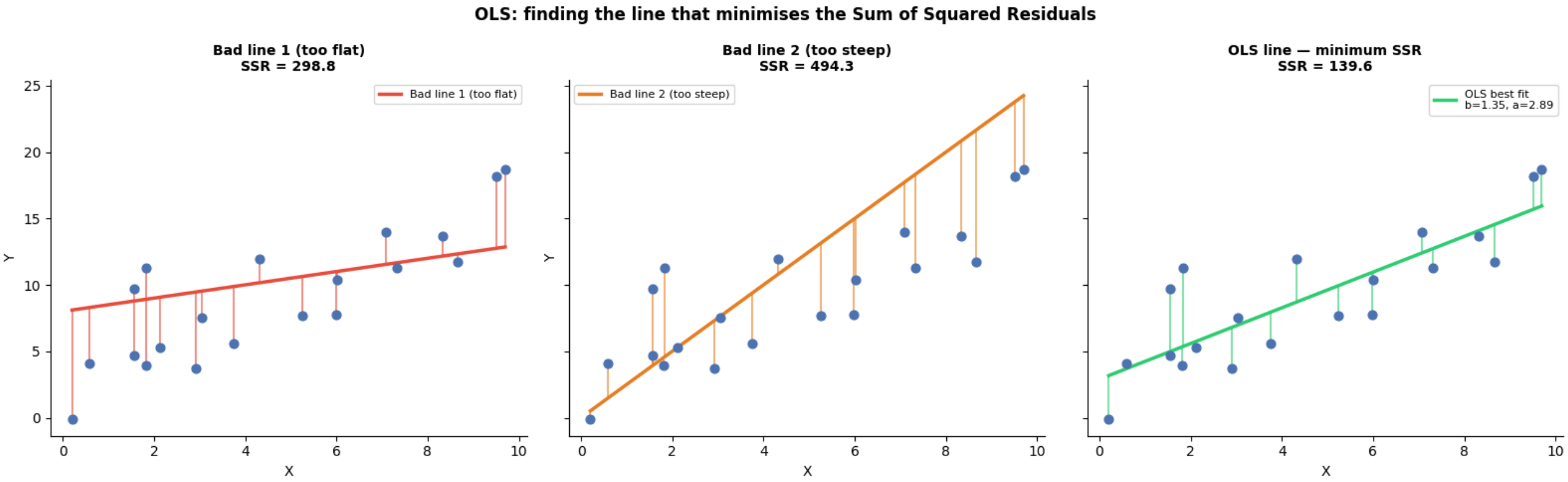
$$S = \frac{\mu - R_f}{\sigma}$$

"Is the return worth the risk?"

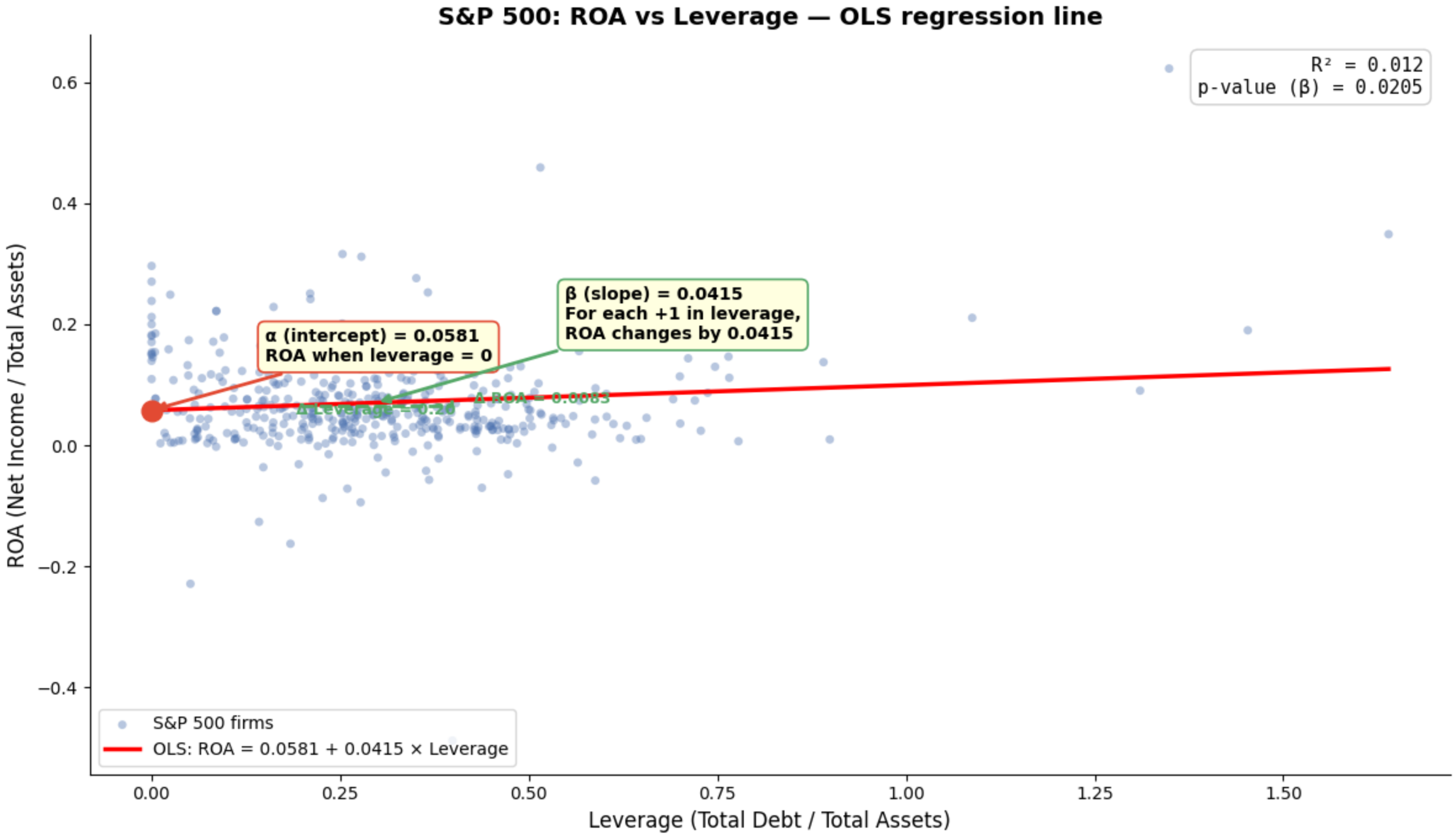
6. Portfolio analysis



7. Regression - univariate



$$SSR = \sum_{i=1}^{n} e_i^2$$


S&P 500: ROA vs Leverage — OLS regression line
R² = 0.012
p-value (β) = 0.0205
α (intercept) = 0.0581
ROA when leverage = 0
β (slope) = 0.0415
For each +1 in leverage,
ROA changes by 0.0415
Δ Leverage = 0.20
Δ ROA = 0.0083
S&P 500 firms
OLS: ROA = 0.0581 + 0.0415 × Leverage
ROA (Net Income / Total Assets)
Leverage (Total Debt / Total Assets)
0.6
0.4
0.2
0.0
−0.2
−0.4
0.00
0.25
0.50
0.75
1.00
1.25
1.50

7. Regression - multivariate

$$ROA_i = \alpha + \beta_1 \cdot Leverage_i + \beta_2 \cdot ESG_i + \varepsilon_i$$

The coefficient on *Leverage* in the regression *ROA* on *Leverage* + *ESG* is the same as:

- The coefficient on *Leverage* in the regression of ROA on *Residual Leverage*.
- Whereby *Residual Leverage* is the residual of the regression of *Leverage* on *ESG* (it captures the variation of *Leverage* not associated with variation in *ESG*).

7. Regression - multivariate

```
======================================================================
MODEL A: ROA ~ Leverage + ESG  (multiple regression)
======================================================================
                            OLS Regression Results
==============================================================================
Dep. Variable:                    roa   R-squared:                       0.015
Model:                            OLS   Adj. R-squared:                  0.011
Method:                 Least Squares   F-statistic:                     3.334
Date:                Fri, 27 Feb 2026   Prob (F-statistic):             0.0366
Time:                        11:40:20   Log-Likelihood:                 482.66
No. Observations:                 428   AIC:                            -959.3
Df Residuals:                     425   BIC:                            -947.1
Df Model:                           2
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0835      0.020      4.213      0.000       0.045       0.123
leverage       0.0414      0.018      2.291      0.022       0.006       0.077
esg           -0.0089      0.007     -1.315      0.189      -0.022       0.004
==============================================================================
Omnibus:                      103.748   Durbin-Watson:                   2.055
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             2274.975
Skew:                           0.396   Prob(JB):                         0.00
Kurtosis:                      14.267   Cond. No.                         17.5
==============================================================================
```

Coef on *Leverage* is 0.04
(p-value = 0.022)

7. Regression - multivariate

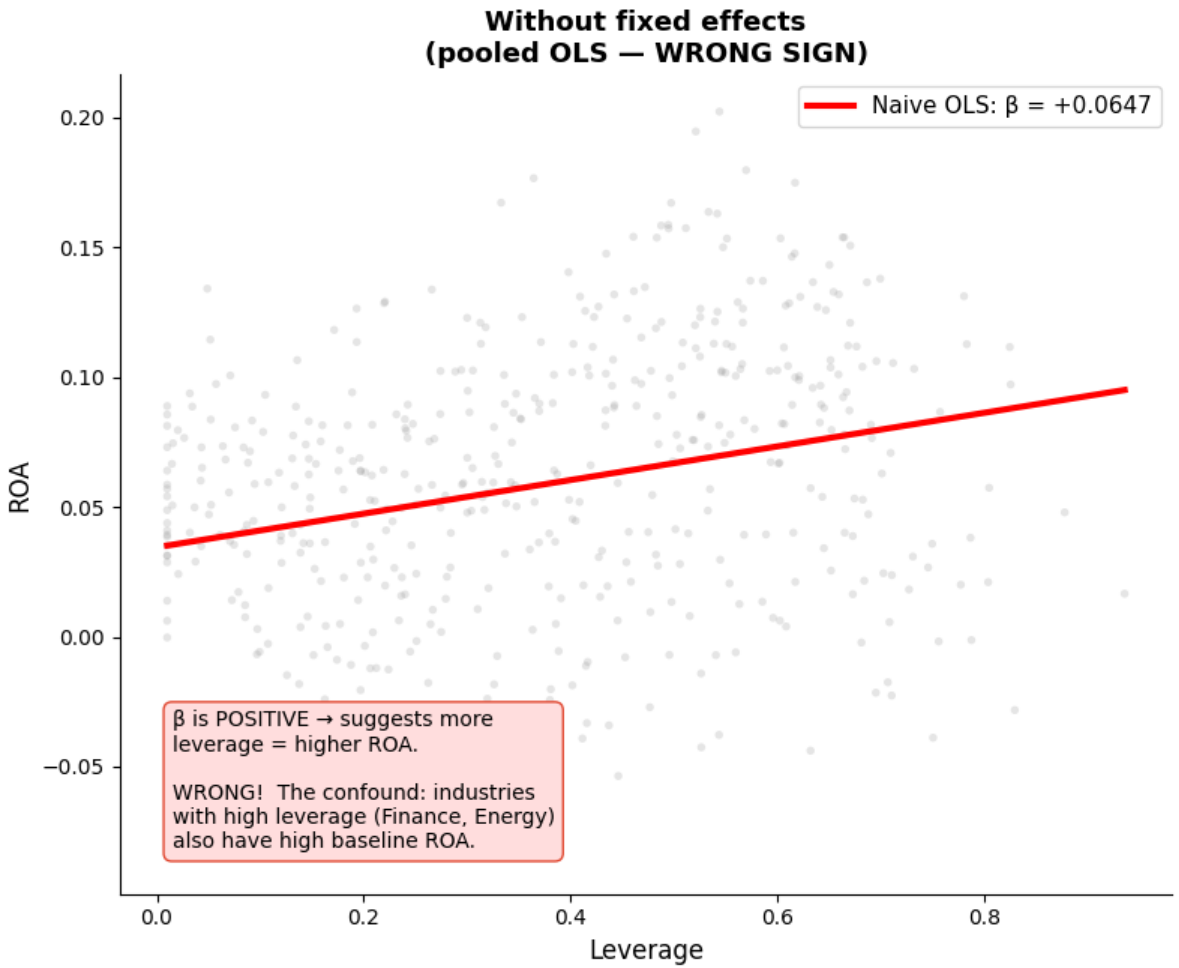
```
==================================================================
MODEL B (Step 1): Leverage ~ ESG  (auxiliary regression)
==================================================================
                            OLS Regression Results
==============================================================================
Dep. Variable:               leverage   R-squared:                       0.003
Model:                            OLS   Adj. R-squared:                  0.001
Method:                 Least Squares   F-statistic:                     1.262
Date:                Fri, 27 Feb 2026   Prob (F-statistic):              0.262
Time:                        11:40:20   Log-Likelihood:                 59.696
No. Observations:                 428   AIC:                            -115.4
Df Residuals:                     426   BIC:                            -107.3
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.2457      0.052      4.737      0.000       0.144       0.348
esg            0.0203      0.018      1.123      0.262      -0.015       0.056
==============================================================================
Omnibus:                      196.476   Durbin-Watson:                   2.037
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             1304.103
Skew:                           1.856   Prob(JB):                    6.57e-284
Kurtosis:                      10.704   Cond. No.                         16.3
==============================================================================
```

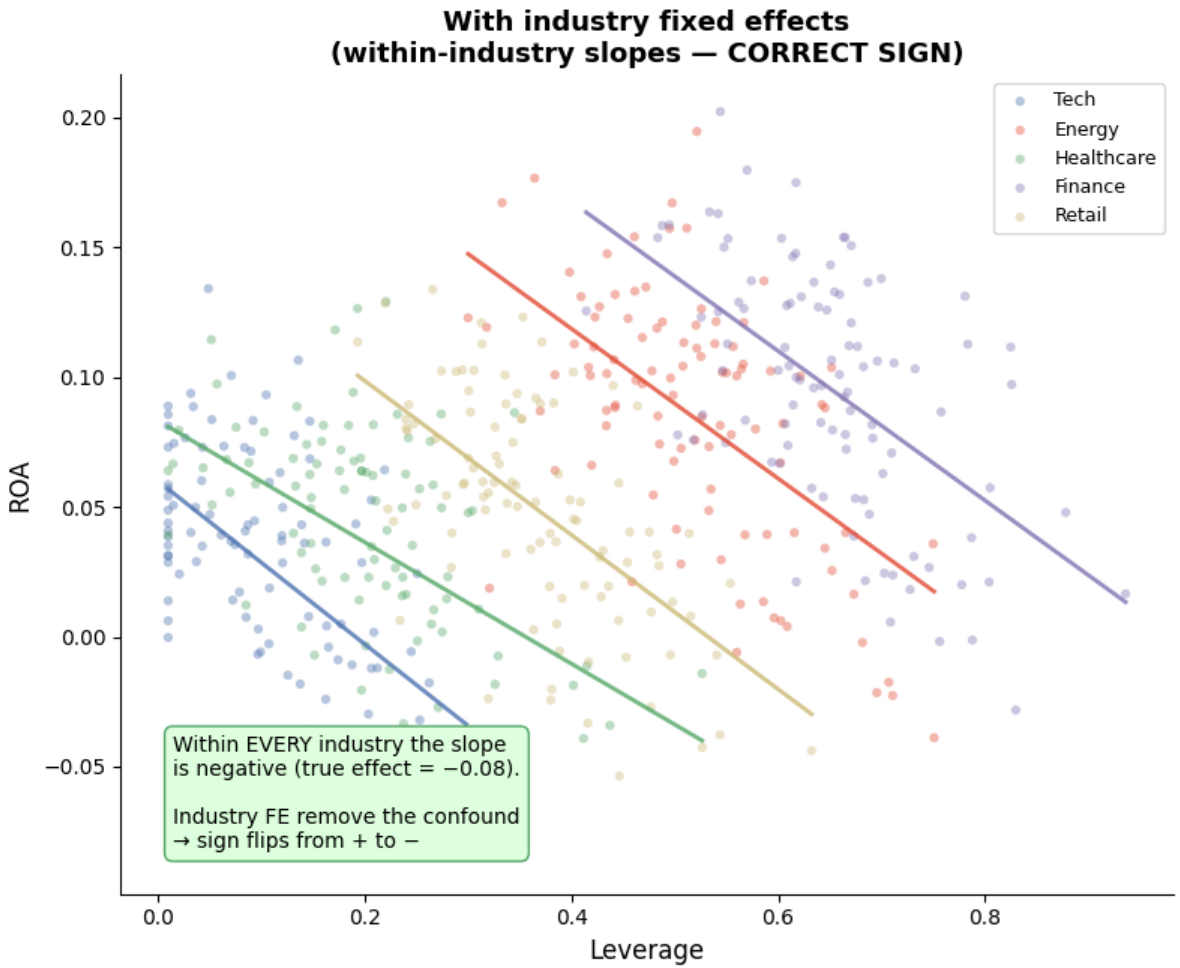
```
==================================================================
MODEL B (Step 2): ROA ~ Residual Leverage
==================================================================
                            OLS Regression Results
==============================================================================
Dep. Variable:                    roa   R-squared:                       0.012
Model:                            OLS   Adj. R-squared:                  0.010
Method:                 Least Squares   F-statistic:                     5.242
Date:                Fri, 27 Feb 2026   Prob (F-statistic):             0.0225
Time:                        11:40:20   Log-Likelihood:                 481.95
No. Observations:                 428   AIC:                            -959.9
Df Residuals:                     426   BIC:                            -951.8
Df Model:                           1
Covariance Type:            nonrobust
==================================================================================
                     coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------
const              0.0711      0.004     18.701      0.000       0.064       0.079
resid_leverage     0.0414      0.018      2.290      0.023       0.006       0.077
==================================================================================
Omnibus:                      106.686   Durbin-Watson:                   2.057
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             2311.905
Skew:                           0.444   Prob(JB):                         0.00
Kurtosis:                      14.351   Cond. No.                         4.75
==================================================================================
```

Residual Leverage = Leverage − (0.2457 + 0.0203 × ESG)

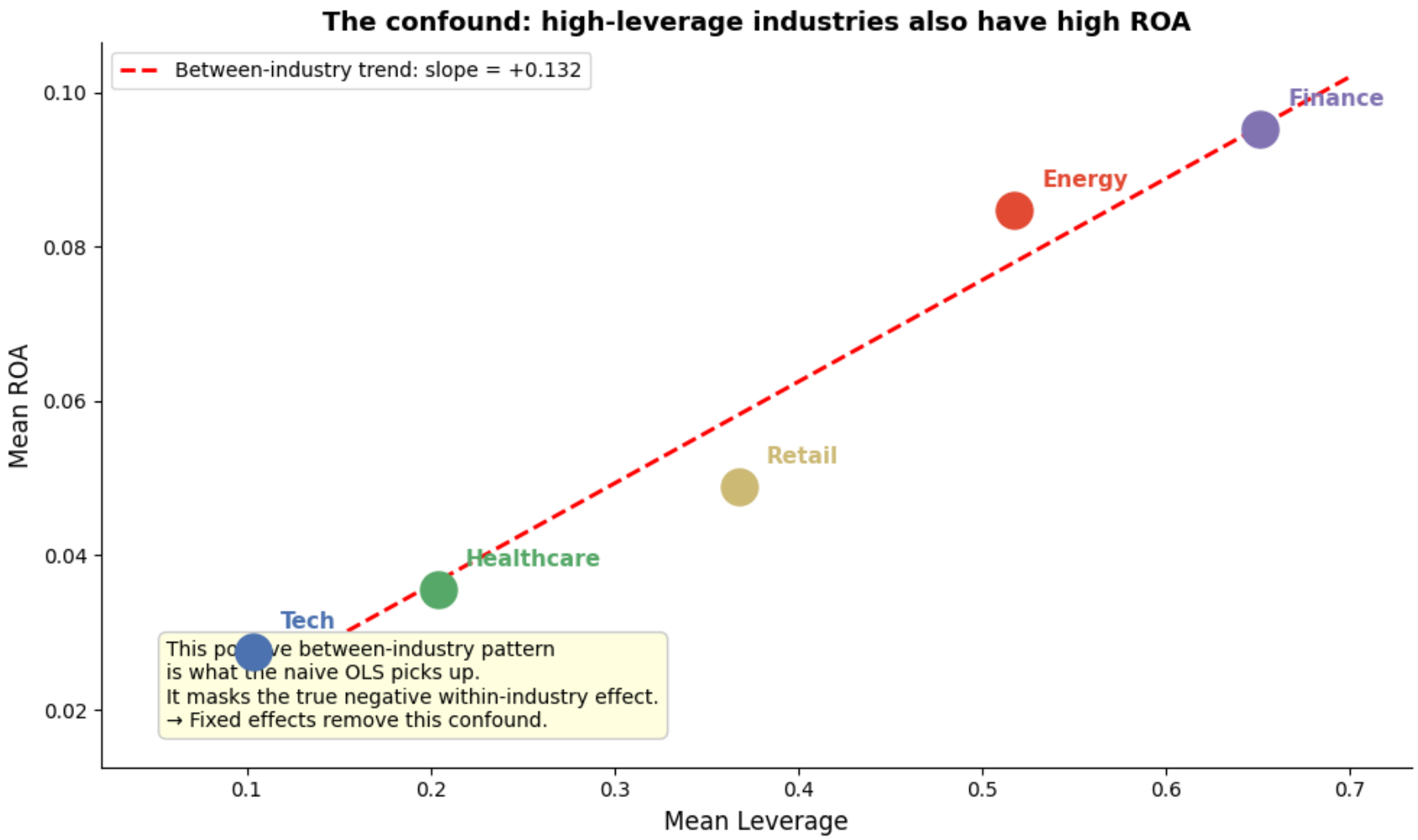
Coef on Residual Leverage is 0.04 (p-value = 0.022)

7. Regression – fixed effects

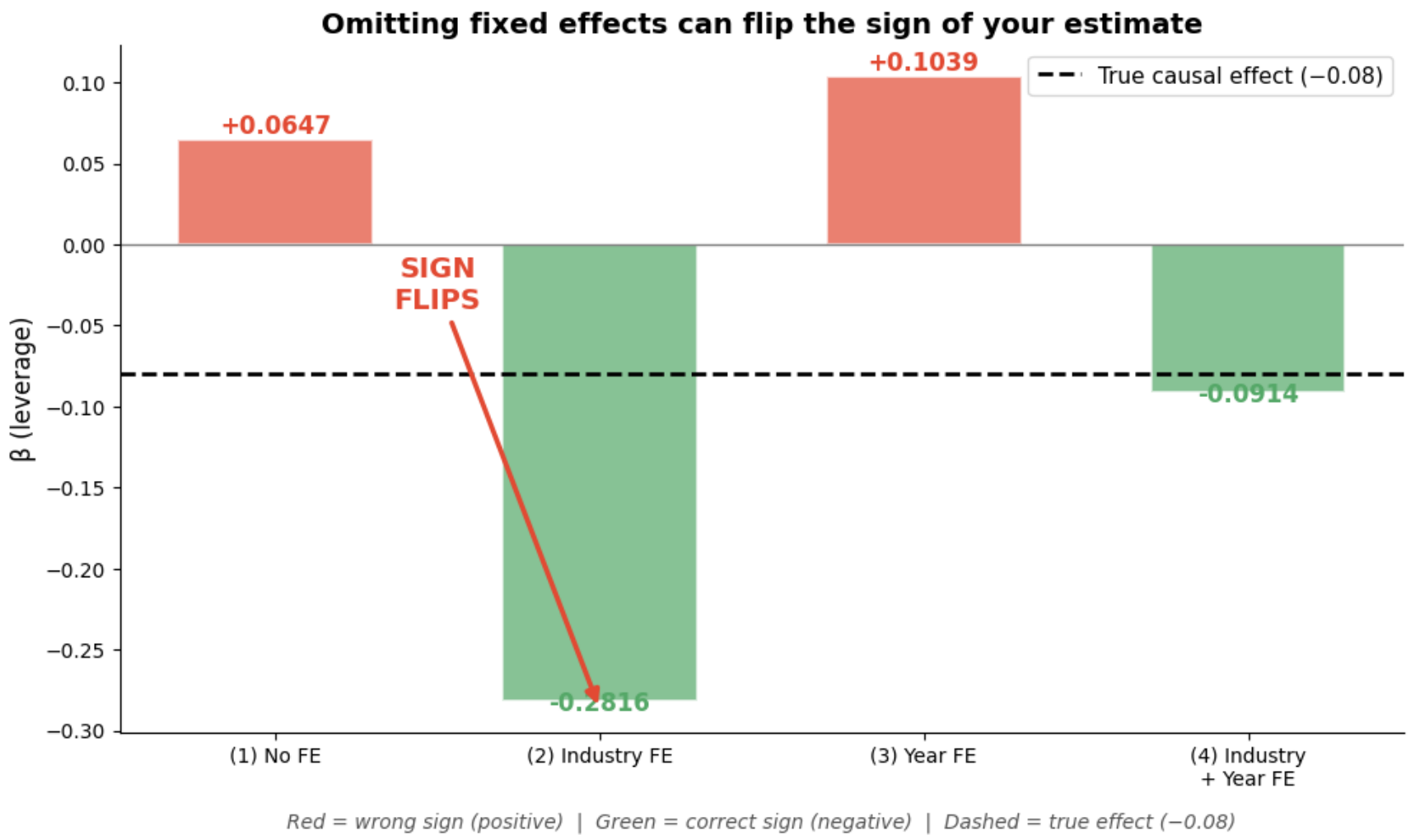




7. Regression – fixed effects



7. Regression – fixed effects



8. Event study

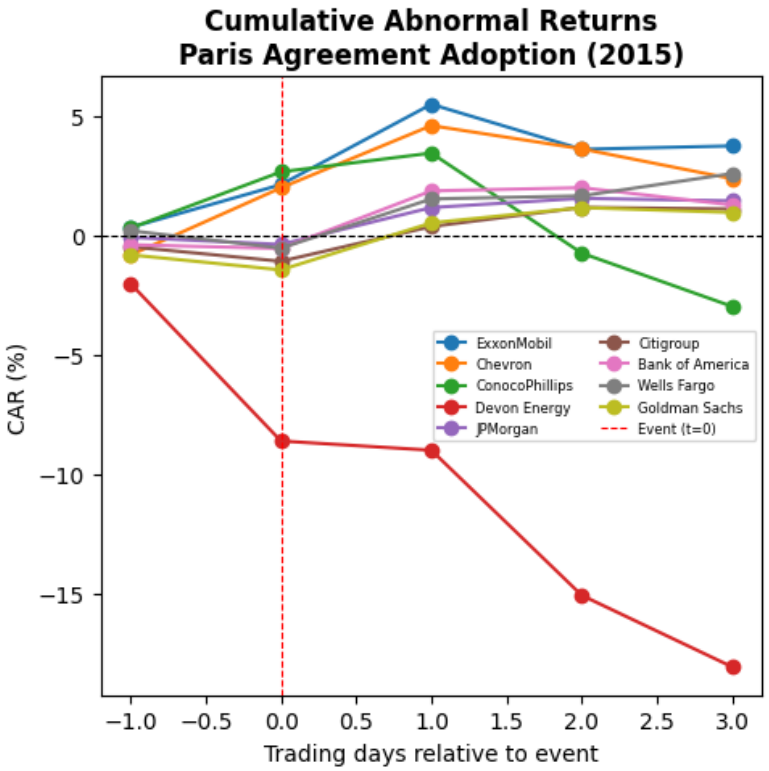
<u>Abnormal reaction of a stock to an event:</u>

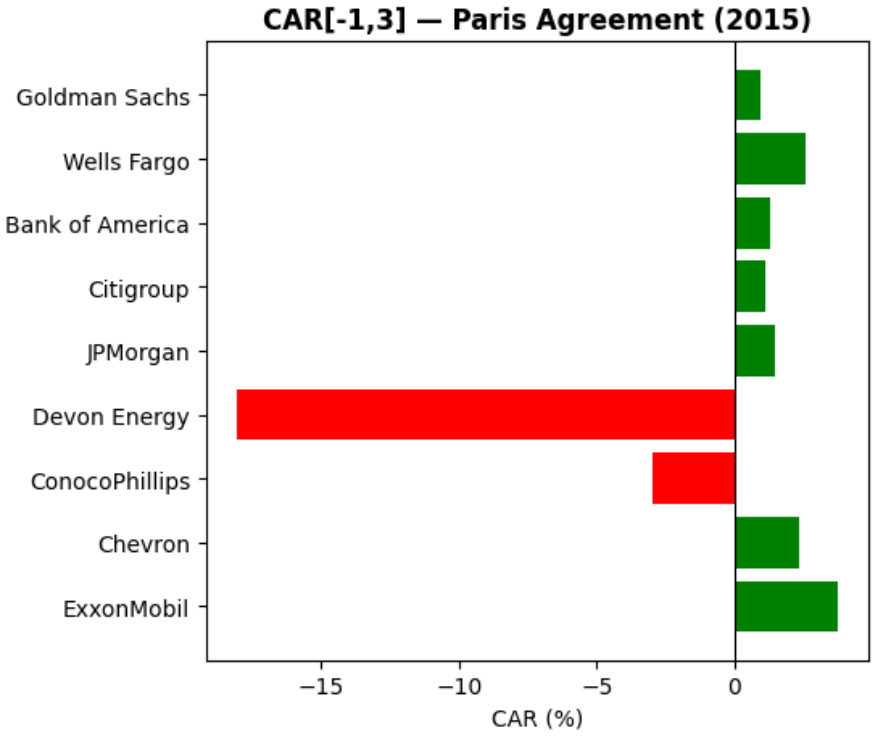
STEP 1. Collect data

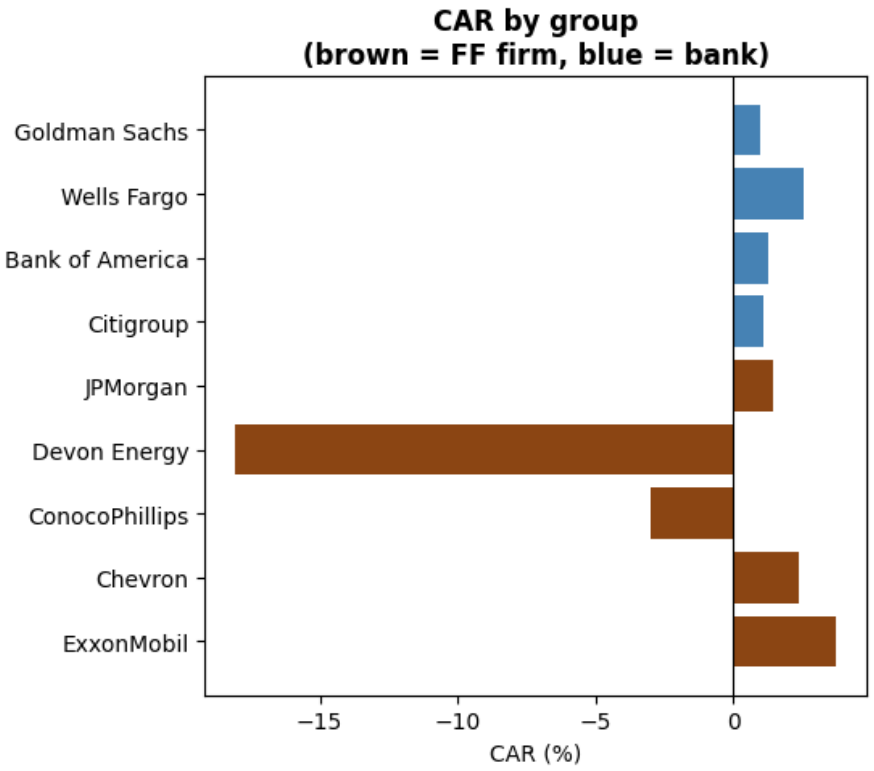
STEP 2. Compute abnormal returns (in excess of market model)

STEP 3. Cumulate abnormal returns over an event window

8. Event study







9. Cursor

See in class demo